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Digital Communication

Course Outcomes

At the end of successful completion of the course the student will be able to

1. Apply concept of sampling for reliable communication.
2. Apply information theory in source coding
3. Analyze distortion less baseband transmission and reception of signal along with waveform coding techniques
4. Analyze different band pass digital transmission and reception of signal.
5. Apply different methods of error control coding and schemes required for reliable transmission of digital communication.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Haykin Simon	<i>Digital Communication Systems,</i>	<i>John Wiley and Sons, India</i>	1 st edition 2013
1.	Taub Schilling	<i>Principles of Communication systems</i>	Tata McGraw Hill, India	4 th Edition 2013
1.	T.L.Singal	<i>Digital Communication</i>	McGraw Hill Education , India	1 st Edition 2015
1.	Bernard Sklar and Pabitra Kumar Ray	<i>Digital communications</i>	Pearson Education	2 nd Edition 2014

Fundamentals of Digital Communications

- What is communication?
- Analog to Digital Conversion (A/D)
- Source Coding
- Channel Encoding
- Modulation Techniques

Block Diagram of a communication system

- Information Source
 - The source of data
 - Data could be: human voice, data storage device CD, video etc..
 - Data types:
 - Discrete: Finite set of outcomes “Digital”
 - Continuous : Infinite set of outcomes “Analog”
- Transmitter
 - Converts the source data into a suitable form for transmission through signal processing
 - Data form depends on the channel



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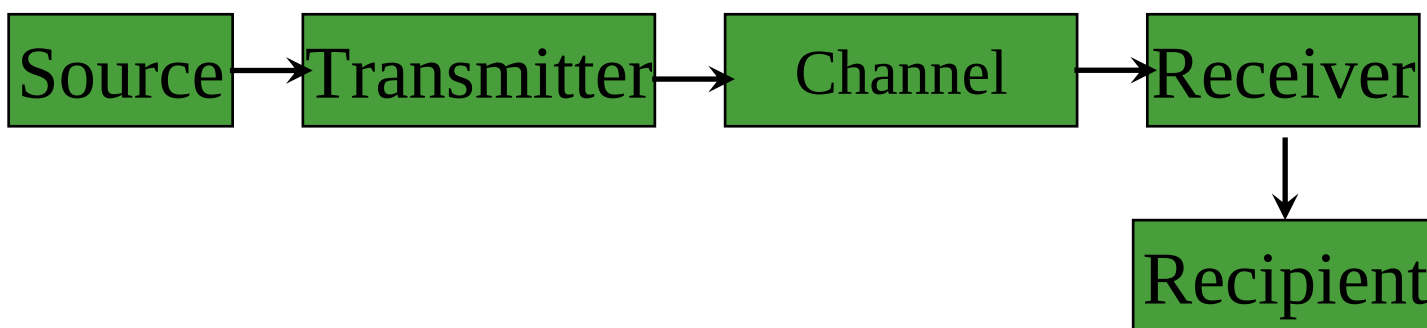


Block Diagram of a communication system

- Main purpose of communication is to transfer information from a source to a recipient via a channel or medium.
- Basic block diagram of a communication system:

Block Diagram of a communication system

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- Basic block diagram of a communication system:



Block Diagram of a communication system

- **Source:** analog or digital
- **Transmitter:** transducer, amplifier, modulator, oscillator, power amp., antenna
- **Channel:** e.g. cable, optical fibre, free space
- **Receiver:** antenna, amplifier, demodulator, oscillator, power amplifier, transducer
- **Recipient:** e.g. person, (loud) speaker, computer

Block Diagram of a communication system

- **Types of information**
Voice, data, video, music, email etc.

- **Types of communication systems**
Public Switched Telephone Network (voice,fax,modem)
Satellite systems
Radio,TV broadcasting
Cellular phones
Computer networks (LANs, WANs, WLANs)

Information Representation

- **Communication system converts information into electrical electromagnetic/optical signals appropriate for the transmission medium.**
- **Analog systems convert analog message into signals that can propagate through the channel.**
- **Digital systems convert bits(digits, symbols) into signals**
 - ❑ Computers naturally generate information as characters/bits
 - ❑ Most information can be converted into bits
 - ❑ Analog signals converted to bits by sampling and quantizing (A/D conversion)

Why digital?

- Digital techniques need to distinguish between discrete symbols allowing regeneration versus amplification
- Good processing techniques are available for digital signals, such as medium.
 - ❑ Data compression (or source coding)
 - ❑ Error Correction (or channel coding)(A/D conversion)
 - ❑ Equalization
 - ❑ Security
- Easy to mix signals and data using digital techniques

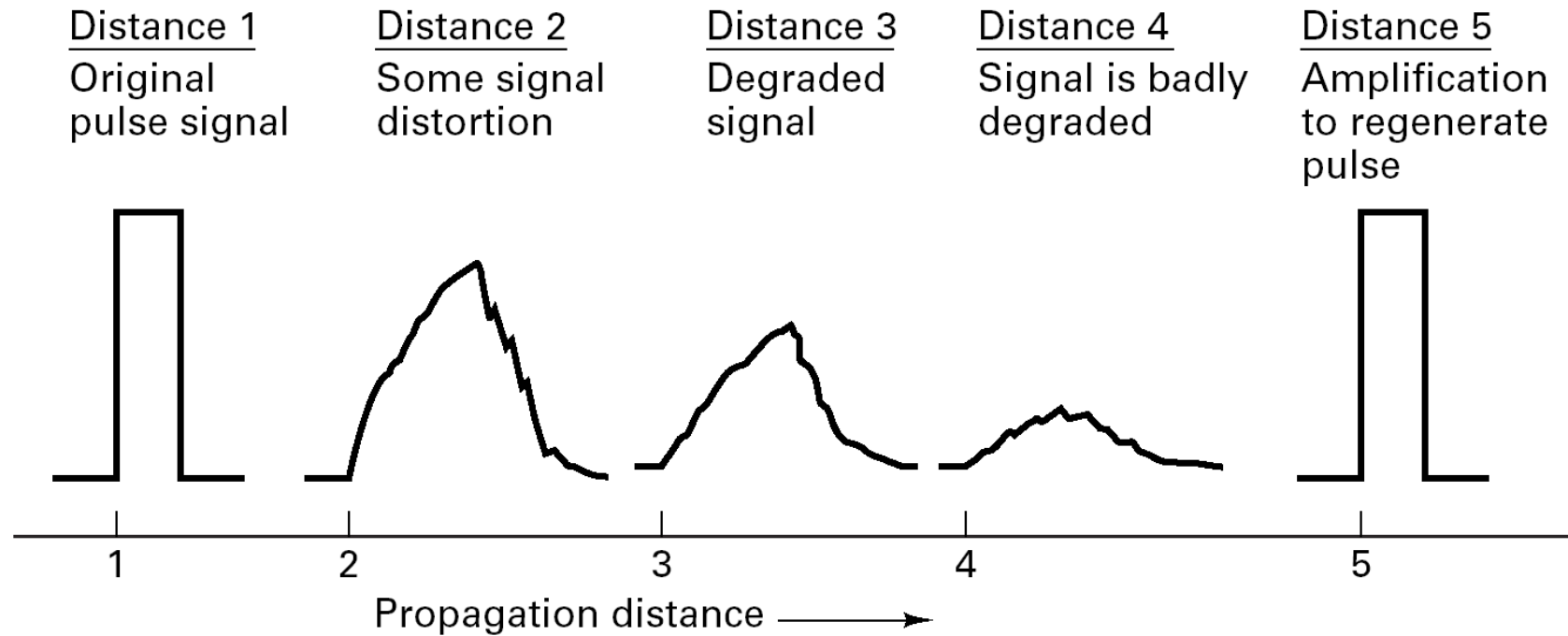


Figure 1.1 Pulse degradation and regeneration.

Block Diagram of a communication system

Channel:

Physical Mediums (Channels):

Wired : example

Wireless: example

channel bandwidth

Physical channels have important limitation: ?

Block Diagram of a communication system

Channel:

Noise

Attenuation

Delay distortion

Receiver:

Extracting the code from received signal

Information sink:

The final stage

The user

Block Diagram of a digital communication system

Information source:

Analog to Digital converter

Sampling

Quantization

Coding

Source encoder

Block Diagram of Digital communication system

Channel encoder

Modulator

Demodulator

Block Diagram of Digital communication system

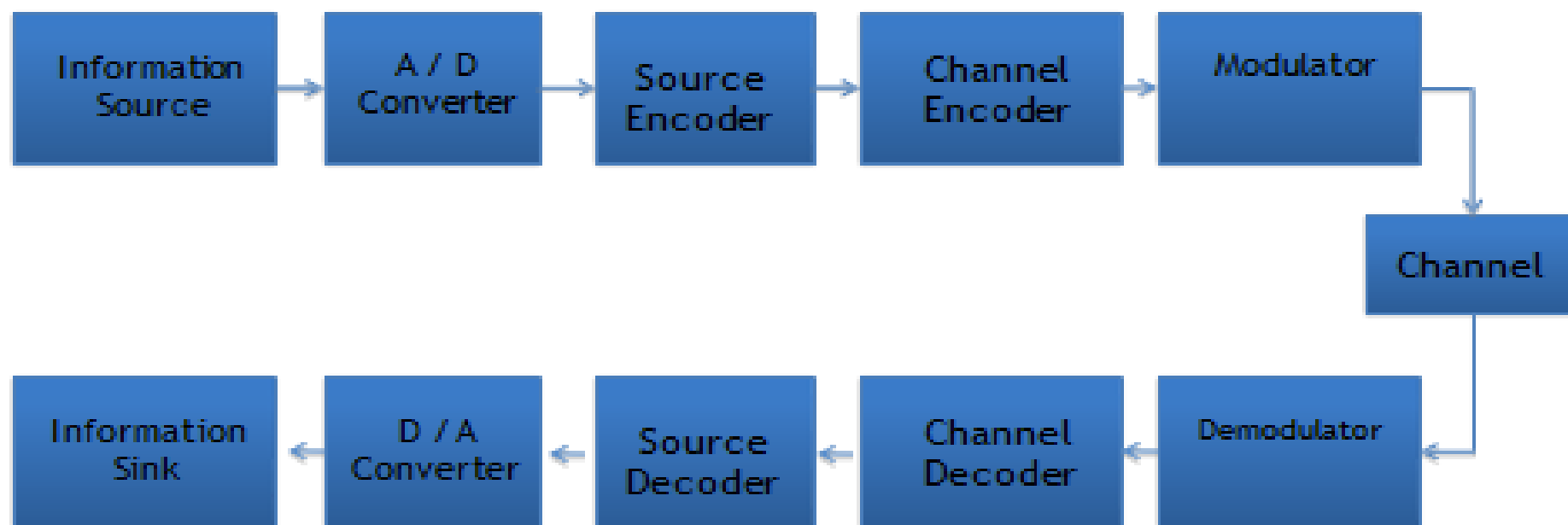
Channel decoder

Source decoder

Digital to Analog converter

Digital Communication System

- Data of a digital format “i.e binary numbers”



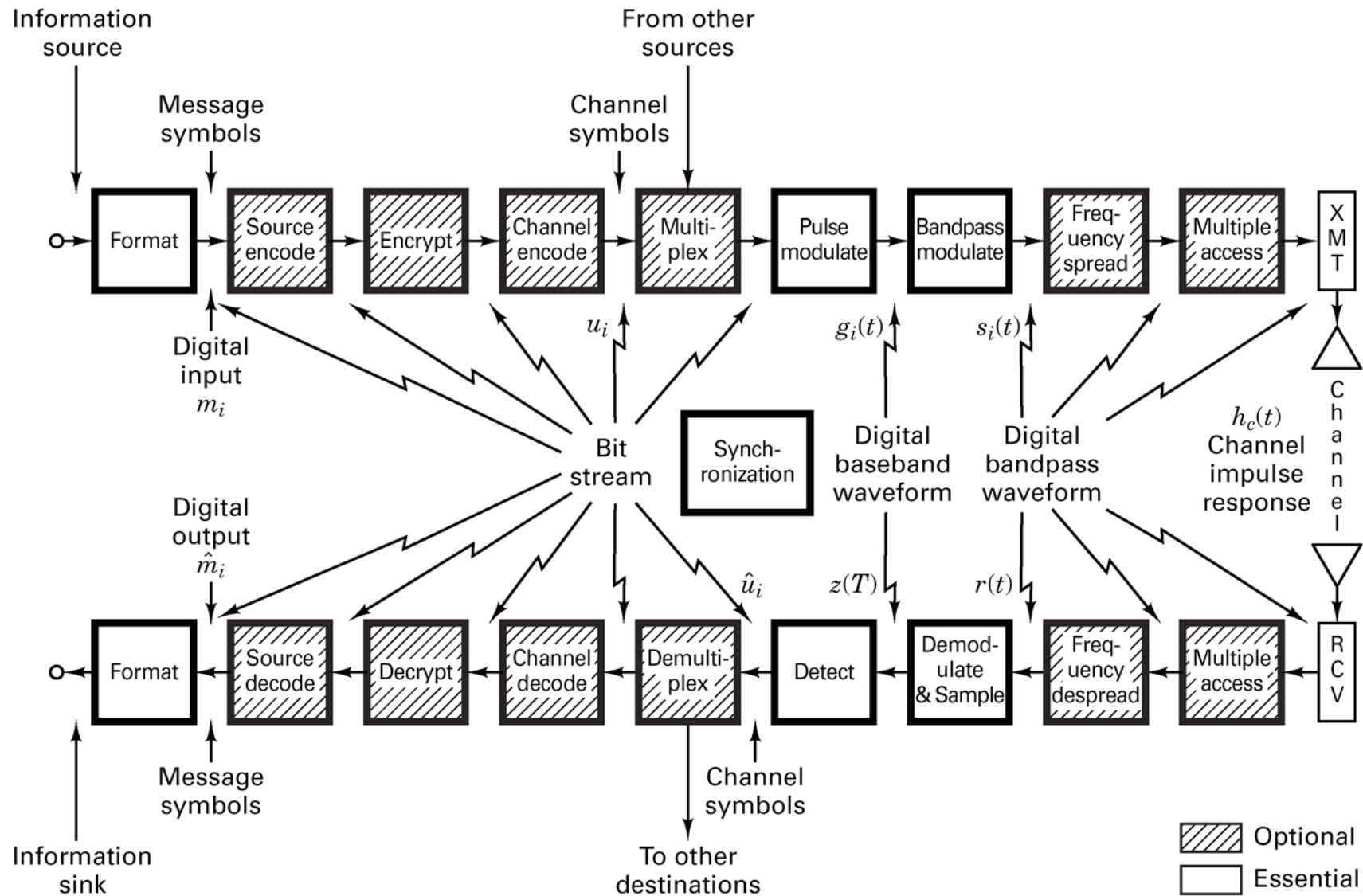


Figure 1.2 Block diagram of a typical digital communication system.

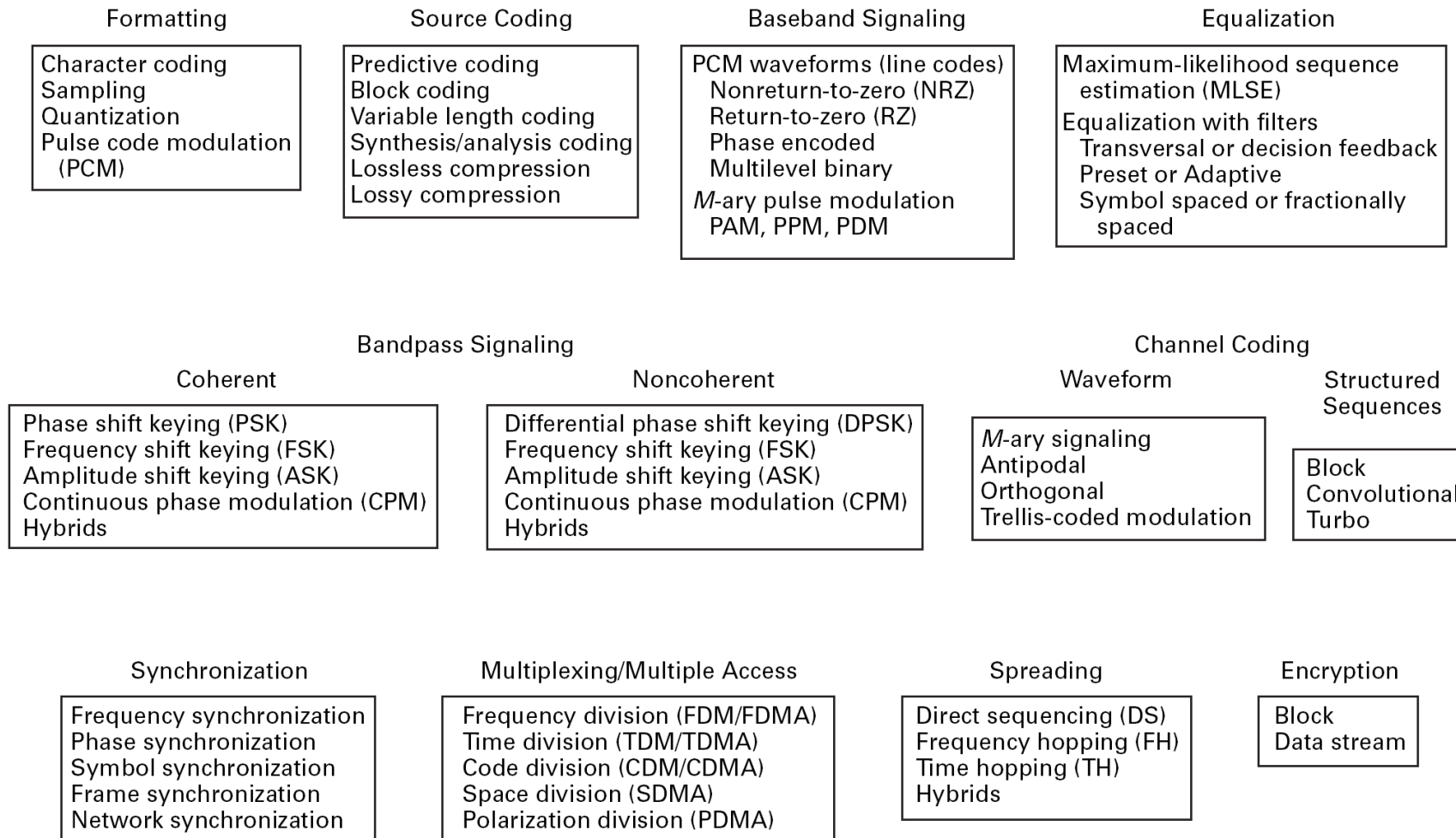


Figure 1.3 Basic digital communication transformations.

■ Formatting/Source Coding

- ❑ Transforms source info into digital symbols (digitization)
- ❑ Selects compatible waveforms (matching function)
- ❑ Introduces redundancy which facilitates accurate decoding despite errors

■ It is essential for reliable communication

- ❑ Modulation/Demodulation
- ❑ Modulation is the process of modifying the info signal to facilitate transmission
- ❑ Demodulation reverses the process of modulation. It involves the detection and retrieval of the info signal
 - Types
 - Coherent: Requires a reference info for detection
 - Noncoherent: Does not require reference phase information

- ❑ Coding/Decoding
 - Translating info bits to transmitter data symbols
 - Techniques used to enhance info signal so that they are less vulnerable to channel impairment (e.g. noise, fading, jamming, interference)
 - Two Categories
 - ❑ *Waveform Coding*
 - Produces new waveforms with better performance
 - ❑ *Structured Sequences*
- Involves the use of redundant bits to determine the occurrence of error (and sometimes correct it)
 - ❑ Multiplexing/Multiple Access Is synonymous with resource sharing with other users
 - ❑ Frequency Division Multiplexing/Multiple Access (FDM/FDMA)

Performance Metrics

Which are the basic performance metrics for digital communication system ? (Advantages)

- Analog Communication Systems
 - ❑ Metric is fidelity: want
 - ❑ SNR typically used as performance metric
- Digital Communication Systems
 - ❑ Metrics are data rate (R bps) and probability of bit error
 - ❑ Symbols already known at the receiver
 - ❑ Without noise/distortion/sync. problem, we will never make bit errors



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Main Points



- Transmitters modulate analog messages or bits in case of a DCS for transmission over a channel.
- Receivers recreate signals or bits from received signal (mitigate channel effects)
- Performance metric for analog systems is fidelity, for digital it is the bit rate and error probability.

Why Digital Communications?

- ❑ Easy to regenerate the distorted signal
- ❑ Regenerative repeaters along the transmission path can detect a digital signal and retransmit a new, clean (noise free) signal
- ❑ These repeaters prevent accumulation of noise along the path
- This is not possible with analog communication systems
 - ❑ Two-state signal representation
- The input to a digital system is in the form of a sequence of bits (binary or M_ary)
 - ❑ Immunity to distortion and interference
 - ❑ Digital communication is rugged in the sense that it is more immune to channel noise and distortion

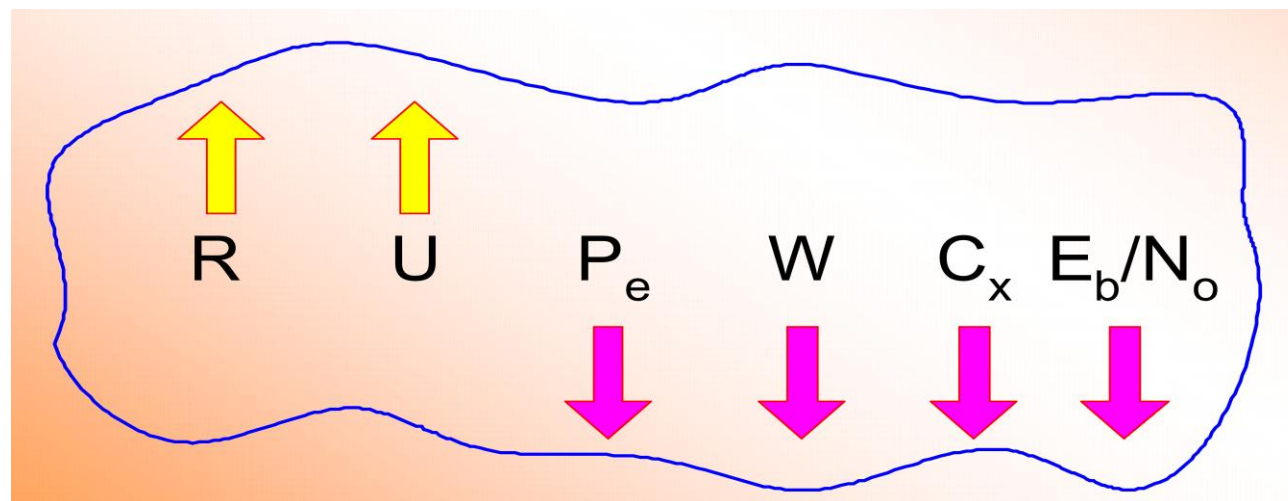
- ❑ Hardware is more flexible
- ❑ Digital hardware implementation is flexible and permits the use of microprocessors, mini-processors, digital switching and VLSI
- Shorter design and production cycle
 - ❑ Low cost
- The use of LSI and VLSI in the design of components and systems have resulted in lower cost
 - ❑ Easier and more efficient to multiplex several digital signals
 - ❑ Digital multiplexing techniques – Time & Code Division Multiple Access - are easier to implement than analog techniques such as Frequency Division Multiple Access

- ❑ Can combine different signal types – data, voice, text, etc.
- ❑ Data communication in computers is digital in nature whereas voice communication between people is analog in nature
- ❑ The two types of communication are difficult to combine over the same medium in the analog domain.
- Using digital techniques, it is possible to combine both format for transmission through a common medium
- Encryption and privacy techniques are easier to implement
 - ❑ Better overall performance
 - ❑ Digital communication is inherently more efficient than analog in realizing the exchange of SNR for bandwidth
 - ❑ Digital signals can be coded to yield extremely low rates and high fidelity as well as privacy

- Requires reliable “synchronization”
- Requires A/D conversions at high rate
- Requires larger bandwidth
- Nongraceful degradation
- Performance Criteria
- Probability of error or Bit Error Rate

Goals in Communication System Design

- To maximize transmission rate, R
- To maximize system utilization, U
- To minimize bit error rate, P_e
- To minimize required systems bandwidth, W
- To minimize system complexity, C_x
- To minimize required power, E_b/N_o



Digital Signal Nomenclature

■ Information Source

- ❑ Discrete output values e.g. Keyboard
- ❑ Analog signal source e.g. output of a microphone

■ Character

- ❑ Member of an alphanumeric/symbol (A to Z, 0 to 9)
- ❑ Characters can be mapped into a sequence of binary digits using **one** of the standardized codes such as
 - **ASCII: American Standard Code for Information Interchange**
 - **EBCDIC: Extended Binary Coded Decimal Interchange Code**

■ Digital Message

- ❑ Messages constructed from a finite number of symbols; e.g., printed language consists of 26 letters, 10 numbers, “space” and several punctuation marks. Hence a text is a digital message constructed from about 50 symbols
- ❑ Morse-coded telegraph message is a digital message constructed from two symbols “**Mark**” and “**Space**”

■ M - ary

- ❑ A digital message constructed with M symbols

■ Digital Waveform

- ❑ Current or voltage waveform that represents a digital symbol

■ Bit Rate

- ❑ Actual rate at which information is transmitted per second

■ Baud Rate

- ❑ Refers to the rate at which the signaling elements are transmitted, i.e. number of signaling elements per second.

■ Bit Error Rate

- ❑ The probability that one of the bits is in error or simply the probability of error

Classification Of Signals

Deterministic and Random Signals

- A signal is *deterministic* means that there is no uncertainty with respect to its value at any time.

- Deterministic waveforms are modeled by explicit mathematical expressions, example:

$$x(t) = 5\cos(10t)$$

- A signal is random means that there is some degree of uncertainty before the signal actually occurs.
- Random waveforms/ Random processes when examined over a long period may exhibit certain regularities that can be described in terms of probabilities and statistical averages.

Periodic and Non-periodic Signals

- A signal $x(t)$ is called *periodic in time* if there exists a constant $T_0 > 0$ such that

$$x(t) = x(t + T_0) \quad \text{for} \quad -\infty < t < \infty$$

t denotes time

T_0 is the *period* of $x(t)$.

Analog and Discrete Signals

- An *analog signal* $x(t)$ is a continuous function of time; that is, $x(t)$ is uniquely defined for all t
- A discrete signal $x(kT)$ is one that exists only at discrete times; it is characterized by a sequence of numbers defined for each time, kT , where
 - k is an integer
 - T is a fixed time interval.

Energy and Power Signals

- The performance of a communication system depends on the received signal *energy*; higher energy signals are detected more reliably (with fewer errors) than are lower energy signals
- $x(t)$ is classified as an energy signal if, and only if, it has nonzero but finite energy ($0 < E_x < \infty$) for all time, where:

$$E_x = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x^2(t) dt = \int_{-\infty}^{\infty} x^2(t) dt$$

- An energy signal **has finite energy but zero average power.**
- Signals that are both deterministic and non-periodic are classified as energy signals

- *Power is the rate at which energy is delivered.*
- *A signal is defined as a power signal if, and only if, it has finite but nonzero power ($0 < P_x < \infty$) for all time, where*

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

- *Power signal **has finite average power but infinite energy.***
- *As a general rule, periodic signals and random signals are classified as power signals*

The Unit Impulse Function

- *Dirac delta function* $\delta(t)$ or impulse function is an abstraction—an infinitely large amplitude pulse, with zero pulse width, and unity weight (area under the pulse), concentrated at the point where its argument is zero.

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\delta(t) = 0 \quad \text{for} \quad t \neq 0$$

$$\delta(t) \text{ is bounded at } t = 0$$

- Sifting or Sampling Property

$$\int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt = x(t_0)$$

Spectral Density

- The *spectral density* of a signal characterizes the distribution of the signal's energy or power in the frequency domain.
- This concept is particularly important when considering filtering in communication systems while evaluating the signal and noise at the filter output.
- The energy spectral density (ESD) or the power spectral density (PSD) is used in the evaluation.

Energy Spectral Density (ESD)

- Energy spectral density describes the signal energy per unit bandwidth measured in joules/hertz.
- Represented as $\psi_x(f)$, the squared magnitude spectrum

$$\psi_x(f) = |X(f)|^2$$

- According to Parseval's theorem, the energy of $x(t)$:

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

- Therefore:

$$E_x = \int_{-\infty}^{\infty} \psi_x(f) df$$

- The Energy spectral density is symmetrical in frequency about origin and total energy of the signal $x(t)$ can be expressed as:

$$E_x = 2 \int_0^{\infty} \psi_x(f) df$$

Power Spectral Density (PSD)

- The power spectral density (PSD) function $G_x(f)$ of the periodic signal $x(t)$ is a real, even, and nonnegative function of frequency that gives the distribution of the power of $x(t)$ in the frequency domain.
- PSD is represented as:

$$G_x(f) = \sum_{n=-\infty}^{\infty} |C_n|^2 \delta(f - nf_0)$$

- Whereas the average power of a periodic signal $x(t)$ is represented as:

$$P_x = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x^2(t) dt = \sum_{n=-\infty}^{\infty} |C_n|^2$$

- Using PSD, the average normalized power of a real-valued signal is represented as:

Autocorrelation of an Energy Signal

- Correlation is a matching process; *autocorrelation* refers to the matching of a signal with a delayed version of itself.
- Autocorrelation function of a real-valued energy signal $x(t)$ is defined as:

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t) x(t + \tau) dt \quad \text{for} \quad -\infty < \tau < \infty$$

- The autocorrelation function $R_x(\tau)$ provides a measure of how closely the signal matches a copy of itself as the copy is shifted τ units in time.
- $R_x(\tau)$ is not a function of time; it is only a function of the time difference τ between the waveform and its shifted copy.

- The autocorrelation function of a real-valued *energy* signal has the following properties:

$$R_x(\tau) = R_x(-\tau)$$

symmetrical in about zero

$$R_x(\tau) \leq R_x(0) \text{ for all } \tau$$

maximum value occurs at the origin

$$R_x(\tau) \leftrightarrow \psi_x(f)$$

autocorrelation and ESD form a Fourier transform pair, as designated by the double-headed arrows

value at the origin is equal to

the energy of the signal

- Autocorrelation function of a real-valued power signal $x(t)$ is defined as:

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t + \tau) dt \quad \text{for } -\infty < \tau < \infty$$

- When the power signal $x(t)$ is periodic with period T_0 , the autocorrelation function can be expressed as

$$R_x(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) x(t + \tau) dt \quad \text{for } -\infty < \tau < \infty$$

- The autocorrelation function of a real-valued *periodic* signal has the following properties similar to those of an energy signal:

$$R_x(\tau) = R_x(-\tau) \quad \text{symmetrical in } \tau \text{ about zero}$$

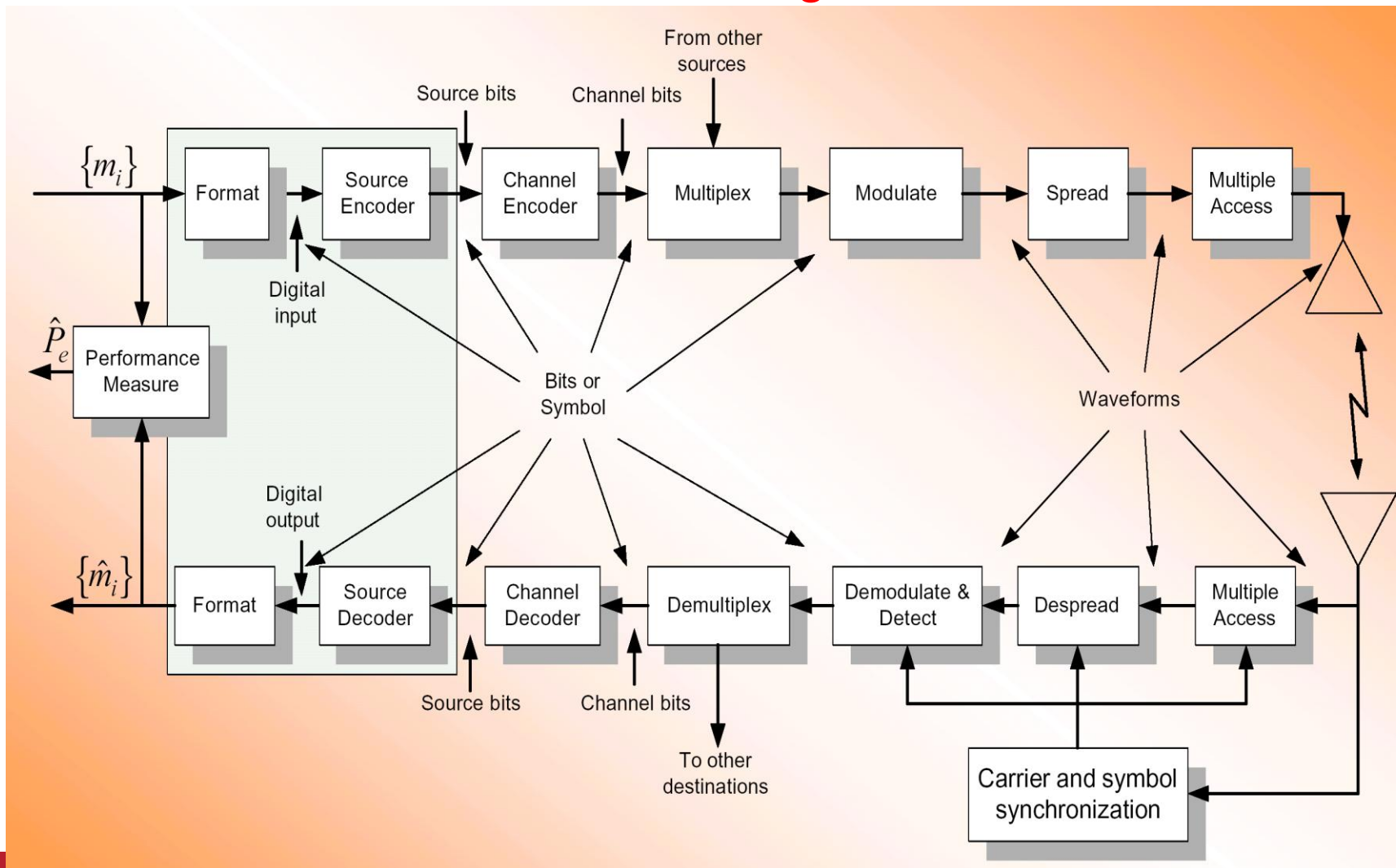
$$R_x(\tau) \leq R_x(0) \text{ for all } \tau \quad \text{maximum value occurs at the origin}$$

$$R_x(\tau) \leftrightarrow G_x(f) \quad \text{autocorrelation and PSD form a Fourier transform pair}$$

$$R_x(0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x^2(t) dt \quad \text{value at the origin is equal to the average power of the signal}$$

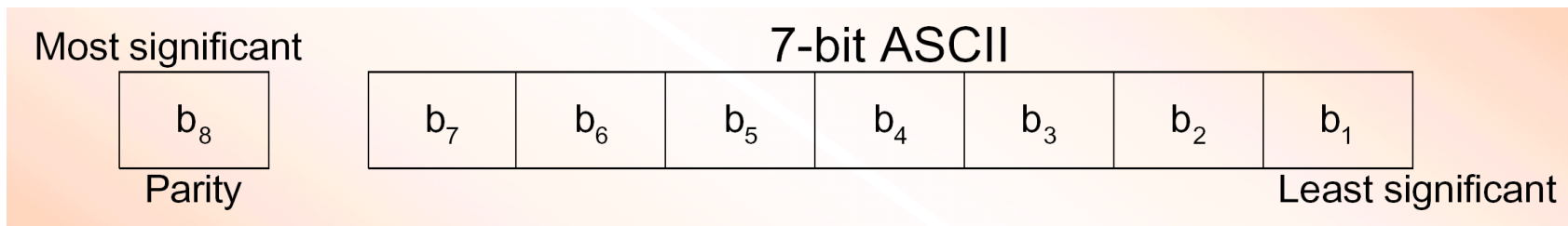
PULSE MODULATION AND DIGITAL TRANSMISSION OF ANALOG SIGNAL

Formatting

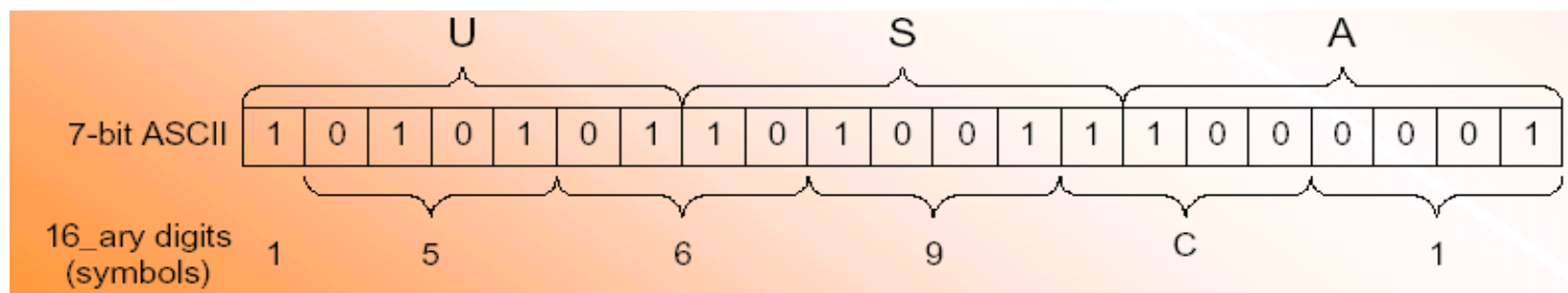


Example 1:

- In ASCII alphabets, numbers, and symbols are encoded using a 7-bit code

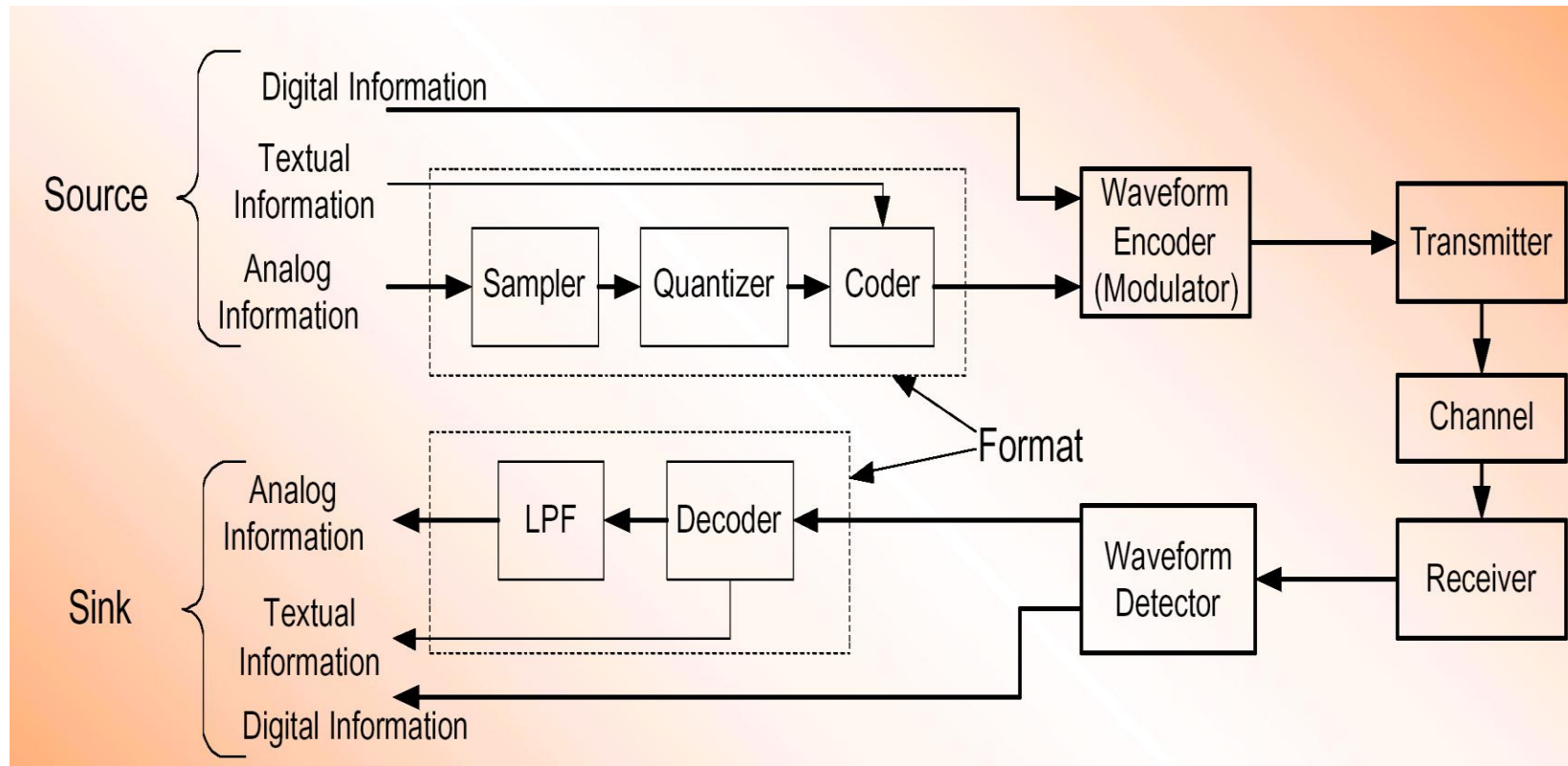


- A total of $2^7 = 128$ different characters can be represented using
- a 7-bit unique ASCII code (see ASCII Table)



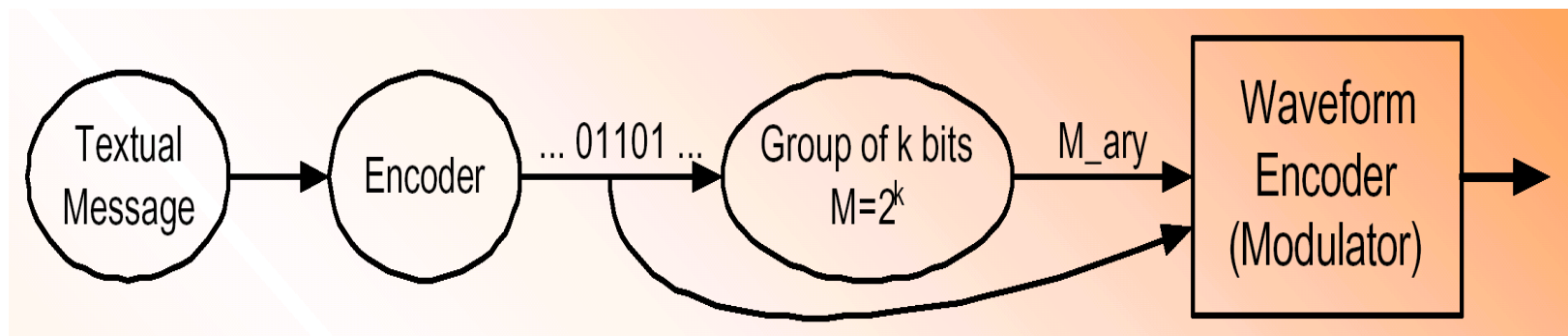
■ Transmit and Receive Formatting

- Transition from *information source* → *digital symbols* → *information sink*



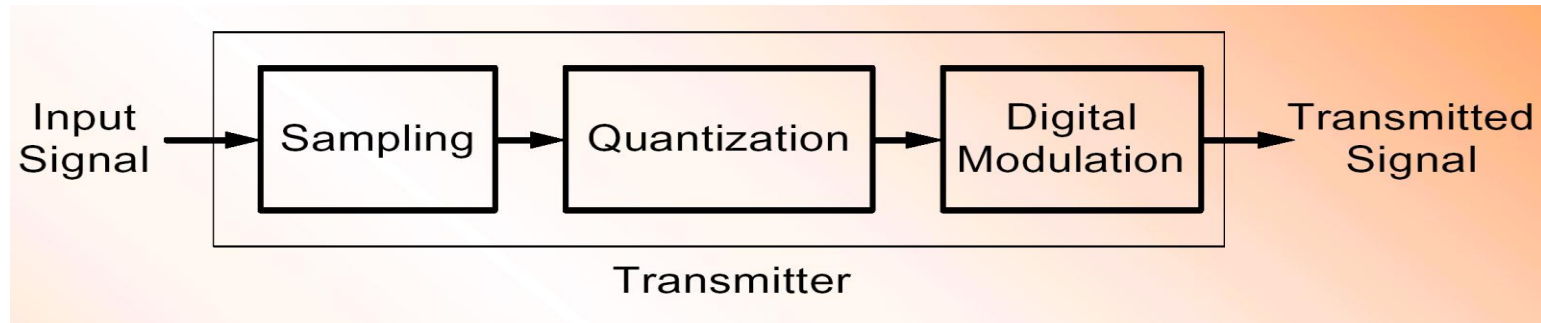
■ Character Coding (Textual Information)

- ❑ A textual information is a sequence of alphanumeric characters
- ❑ Alphanumeric and symbolic information are encoded into digital bits using one of several standard formats, e.g, ASCII, EBCDIC

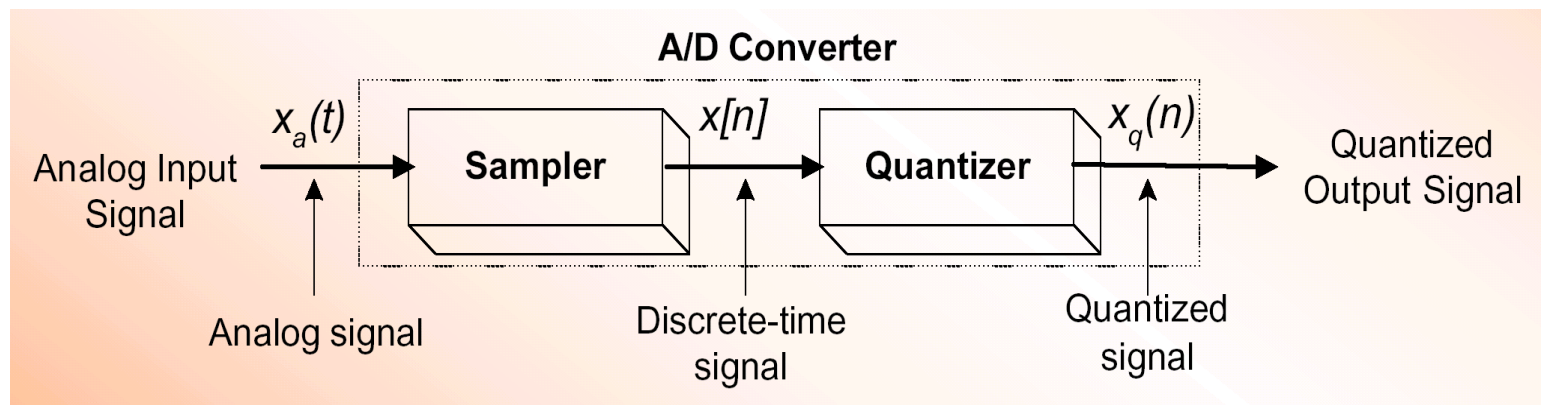


Transmission of Analog Signals

■ Structure of Digital Communication Transmitter



■ Analog to Digital Conversion



Sampling

- **Sampling** is the processes of converting continuous-time analog signal, $x_a(t)$, into a discrete-time signal by taking the “samples” at discrete-time intervals
 - ❑ Sampling analog signals makes them discrete in time but still continuous valued
 - ❑ If done properly (***Nyquist theorem*** is satisfied), sampling does not introduce distortion
- **Sampled values:**
 - ❑ The value of the function at the sampling points
- **Sampling interval:**
 - ❑ The time that separates sampling points (interval b/w samples), T_s
 - ❑ If the signal is slowly varying, then fewer samples per second will be required than if the waveform is rapidly varying
 - ❑ So, the optimum sampling rate depends on the ***maximum frequency*** component present in the signal

- **Analog-to-digital conversion** is (basically) a 2 step process:
 - Sampling
 - Convert from continuous-time analog signal $x_a(t)$ to discrete-time continuous value signal $x(n)$
 - Is obtained by taking the “samples” of $x_a(t)$ at discrete-time intervals, T_s
- **Quantization**
 - Convert from discrete-time continuous valued signal to discrete time discrete valued signal

- **Sampling Rate (or sampling frequency f_s):**

- The rate at which the signal is sampled, expressed as the number of samples per second (reciprocal of the sampling interval), $1/T_s = f_s$

- **Nyquist Sampling Theorem (or Nyquist Criterion):**

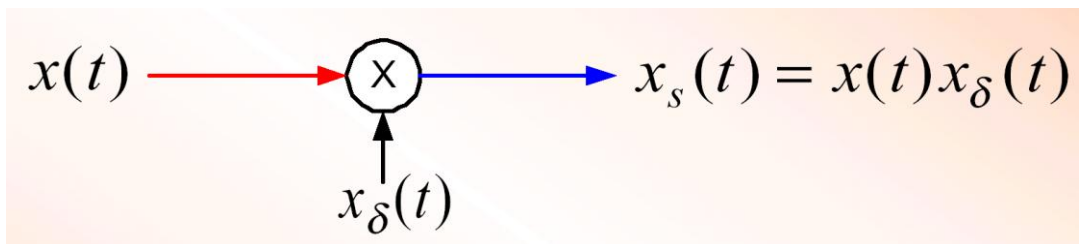
- If the sampling is performed at a proper rate, no info is lost about the original signal and it can be properly reconstructed later on
- Statement:
- “If a signal is sampled at a rate at least, but not exactly equal to twice the max frequency component of the waveform, then the waveform can be exactly reconstructed from the samples without any distortion”

$$f_s \geq 2 f_{\max}$$

- ❑ If $R_s < 2B$, **aliasing** (overlapping of the spectra) results
- ❑ If signal is not strictly bandlimited, then it must be passed through **Low Pass Filter** (LPF) before sampling
- **Fundamental Rule of Sampling (Nyquist Criterion)**
 - ❑ The value of the sampling frequency f_s must be greater than twice the highest signal frequency $f_{\max}$ of the signal
- **Types of sampling**
 - ❑ Ideal Sampling
 - ❑ Natural Sampling
 - ❑ Flat-Top Sampling

Ideal Sampling (or Impulse Sampling)

- Is accomplished by the multiplication of the signal $x(t)$ by the uniform train of impulses (comb function)
- Consider the instantaneous sampling of the analog signal $x(t)$



- Train of impulse functions select sample values at regular intervals

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- Fourier Series representation:

$$\sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn\omega_s t}, \quad \omega_s = \frac{2\pi}{T_s}$$

■ Therefore, we have:

$$x_s(t) = x(t) \left(\frac{1}{T_s} \right) \sum_{n=-\infty}^{\infty} e^{jn\omega_s t}$$

- Take Fourier Transform (frequency convolution)

$$X_s(f) = \frac{1}{T_s} X(f) * \mathfrak{T} \left\{ \sum_{n=-\infty}^{\infty} e^{jn\omega_s t} \right\} = \frac{1}{T_s} X(f) * \sum_{n=-\infty}^{\infty} \mathfrak{T} \{ e^{jn\omega_s t} \}$$

$$X_s(f) = \frac{1}{T_s} X(f) * \sum_{n=-\infty}^{\infty} \delta(f - nf_s), \quad f_s = \frac{\omega_s}{2\pi}$$

$$X_s(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X\left(f - \frac{n}{T_s}\right)$$

Example: 4. Find Inverse F.T of $\delta(\omega - \omega_0)$.

As per defⁿ of I. Fourier x'form,

$$f^{-1}[\delta(\omega - \omega_0)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega \rightarrow \Phi$$

using Sampling property,

$$\int_{-\infty}^{\infty} \phi(t) \delta(t - \tau) dt = \phi(\tau) \quad \text{or} \quad \int_{-\infty}^{\infty} \phi(\omega) \delta(\omega - \omega_0) d\omega = \phi(\omega_0)$$

becomes,

$$= \frac{1}{2\pi} e^{j\omega_0 t}$$

Therefore,

$$\frac{1}{2\pi} e^{j\omega_0 t} \longleftrightarrow \delta(\omega - \omega_0)$$

or

$$e^{j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega - \omega_0)$$

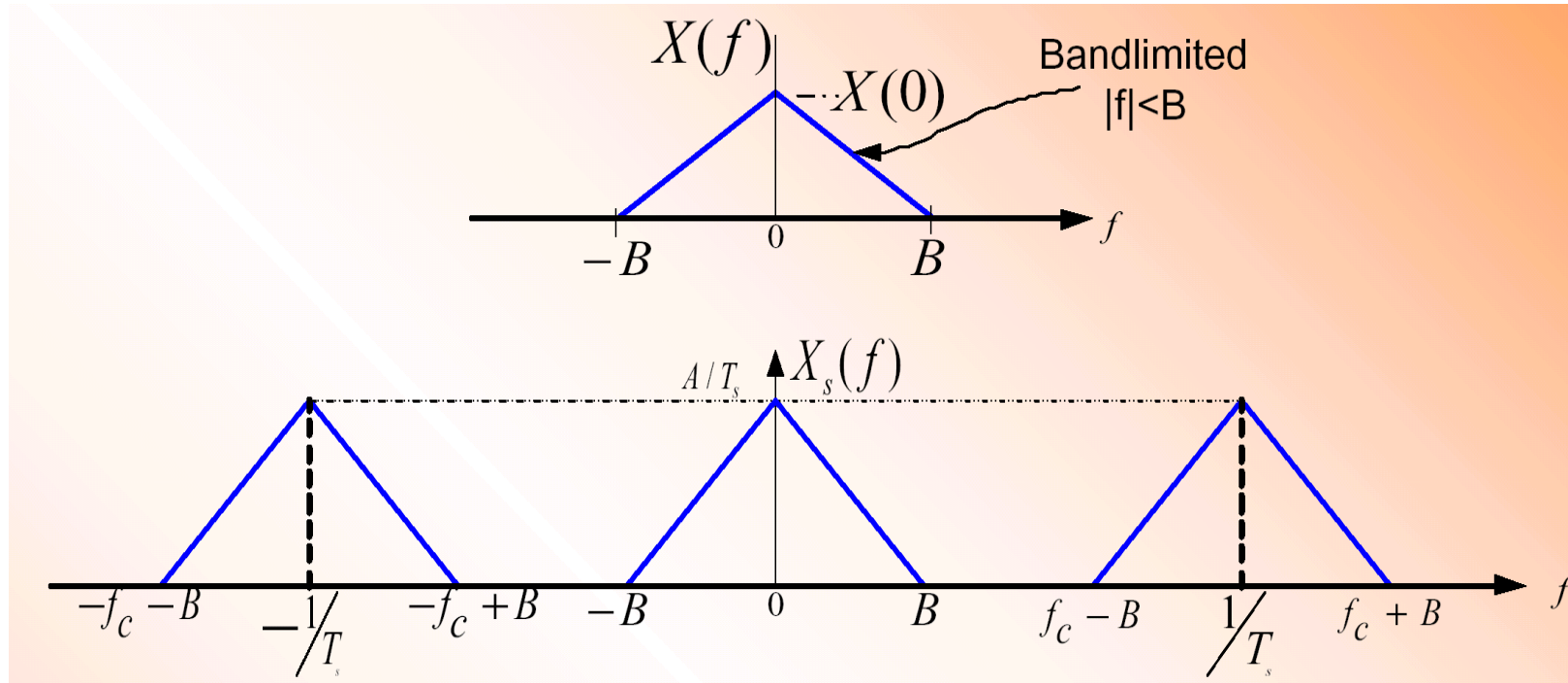
and

from above eqⁿ, it follows that,

$$e^{-j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega + \omega_0)$$

Ideal Sampling (or Impulse Sampling)

- This shows that the Fourier Transform of the sampled signal is the Fourier Transform of the original signal at rate of $1/T_s$



Therefore,

$$g(t) \rightleftharpoons \sum_{n=-\infty}^{\infty} F[D_n e^{jn\omega_0 t}]$$

As we have already derive,

$e^{jn\omega_0 t} \Rightarrow 2\pi \delta(\omega - n\omega_0)$ so, it follows that,

$$g(t) = 2\pi \sum_{n=-\infty}^{\infty} D_n \delta(\omega - n\omega_0) \quad \text{--- (1)}$$

→ impulse train can be expressed as exponential Fourier series as,

$$\delta_{T_0}(t) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t} \quad \omega_0 = \frac{2\pi}{T_0}$$

Here,

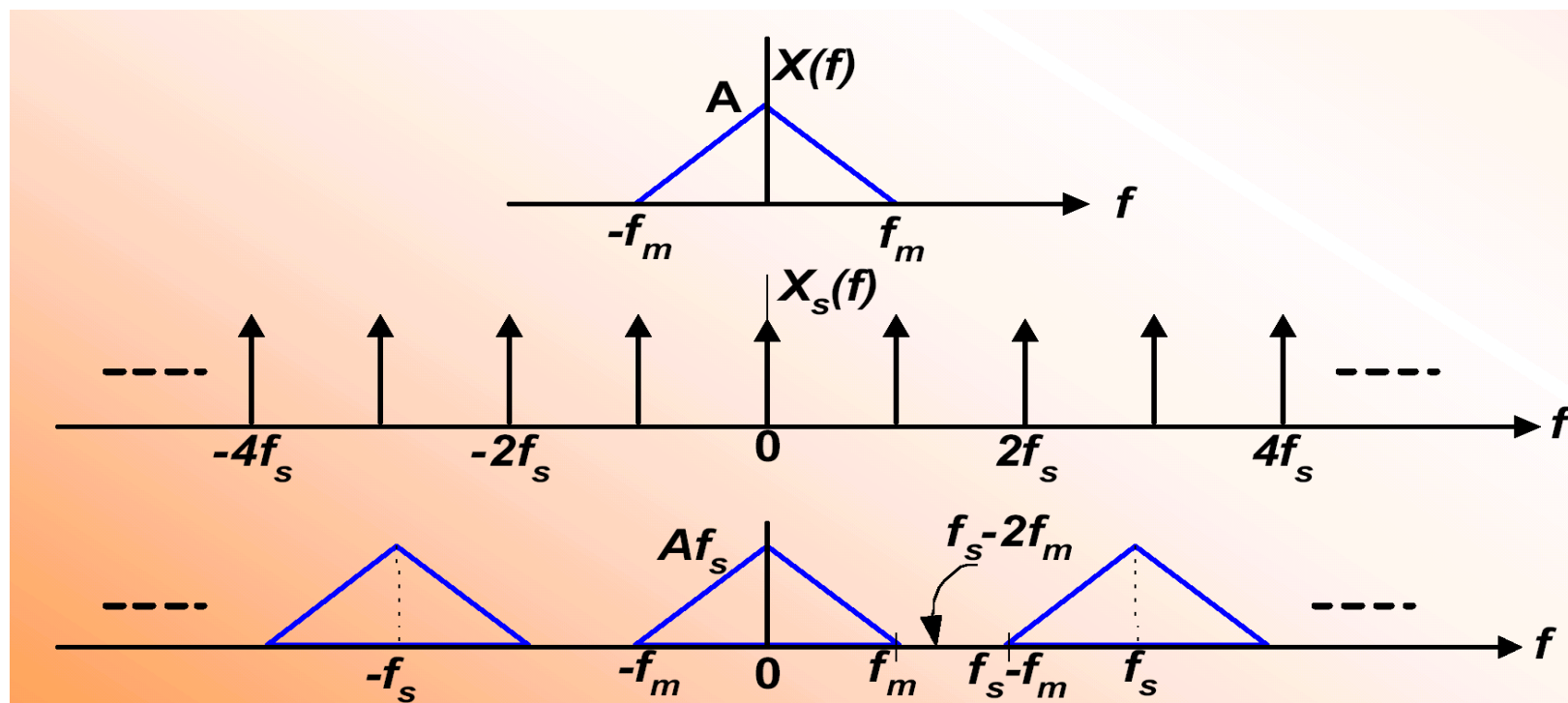
$D_n = 1/T_0$ therefore, from eq (1),

$$\delta_{T_0} \rightleftharpoons \boxed{\frac{2\pi}{T_0} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)}$$

$$\delta_{T_0} \rightleftharpoons \omega_0 \delta_{\omega_0}(\omega) \quad ; \quad \omega_0 = \frac{2\pi}{T_0}$$

Thus, spectrum of impulse train as shown in fig (a) happens to be an impulse train as shown in fig (b).

- This means that the output is simply the replication of the original signal at discrete intervals, e.g



- As long as $f_s > 2f_m$, no overlap of repeated replicas $X(f - n/T_s)$ will occur in $X_s(f)$

$$f_s - f_m > f_m \Rightarrow f_s > 2f_m$$

- Minimum Sampling Condition:

- **Sampling Theorem:** A finite energy function $x(t)$ can be completely **reconstructed** from its sampled value $x(nT_s)$ with

$$x(t) = \sum_{n=-\infty}^{\infty} T_s x(nT_s) \left\{ \frac{\sin \left[\frac{2\pi f(t - nT_s)}{2T_s} \right]}{\pi(t - nT_s)} \right\}$$

$$= \sum_{n=-\infty}^{\infty} T_s x(nT_s) \operatorname{sinc}(2f_s(t - nT_s))$$

provided that $\Rightarrow \frac{1}{f_s} = T_s \leq \frac{1}{2f_m}$

T_s is called the **Nyquist interval**: It is the longest time interval that can be used for sampling a bandlimited signal and still allow reconstruction of the signal at the receiver without distortion

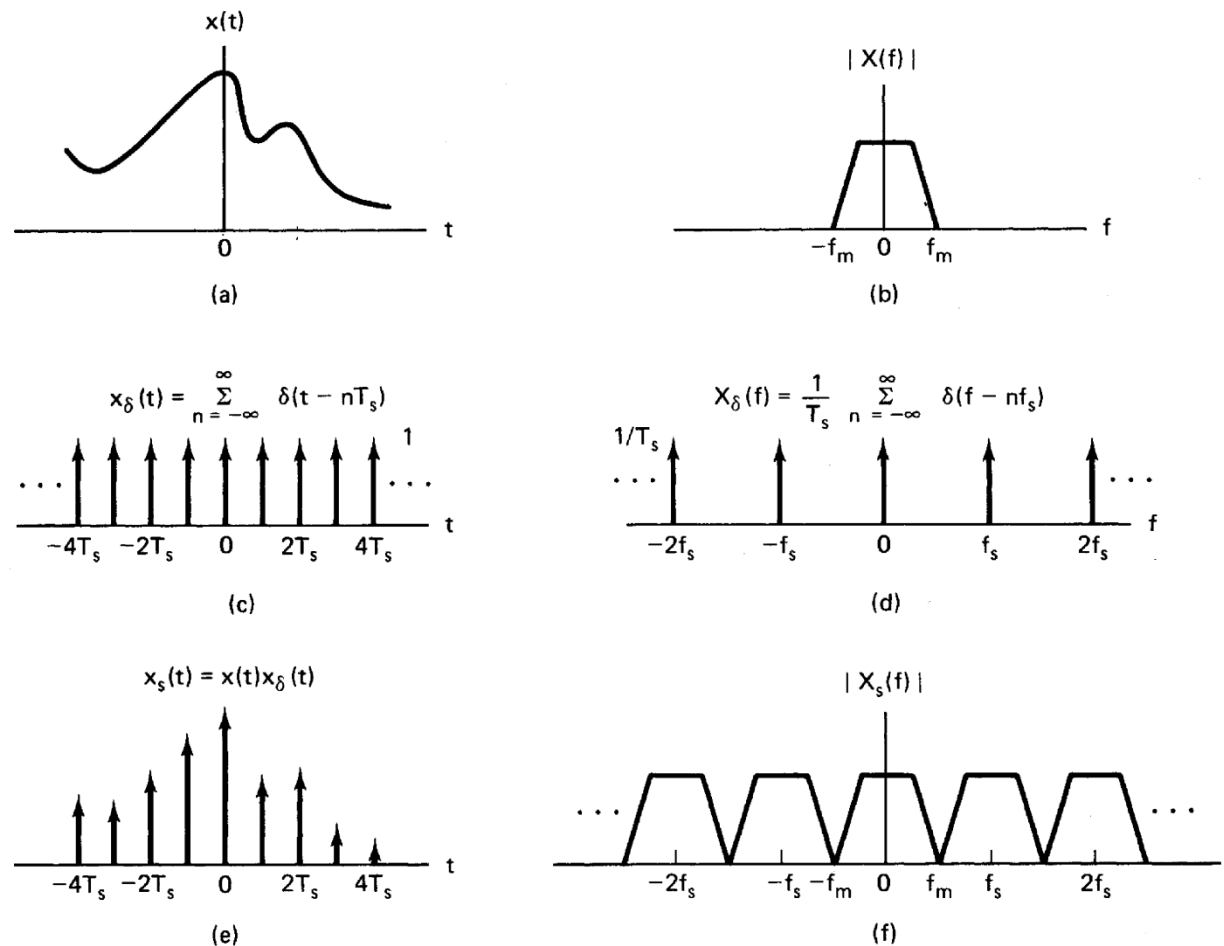
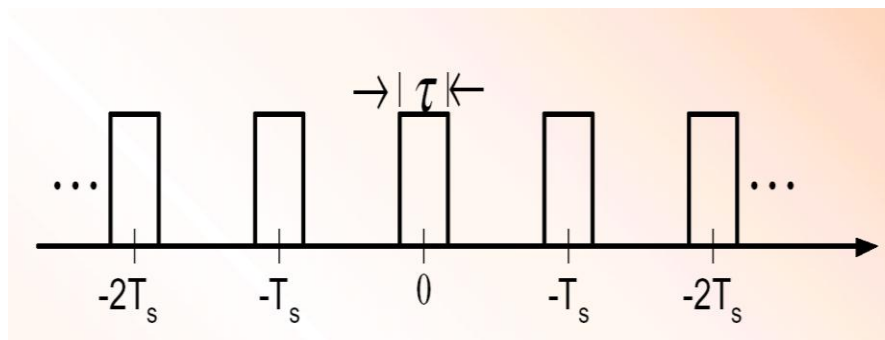


Figure 2.6 Sampling theorem using the frequency convolution property of the Fourier transform.

- In practice we cannot perform ideal sampling
 - It is practically difficult to create a train of impulses
- Thus a non-ideal approach to sampling must be used
- We can approximate a train of impulses using a train of very thin rectangular pulses:

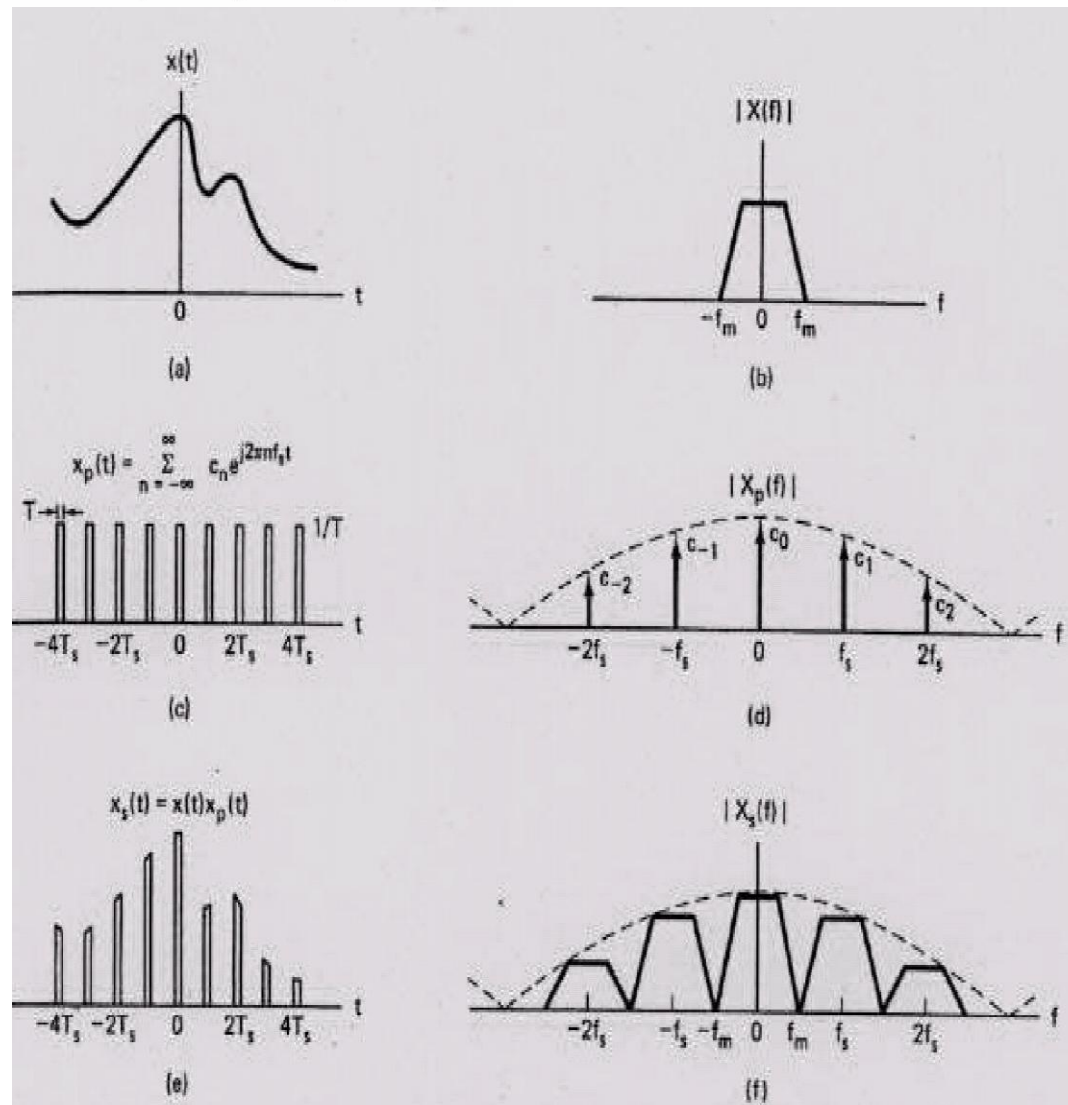
$$x_p(t) = \sum_{n=-\infty}^{\infty} \Pi\left(\frac{t - nT_s}{\tau}\right)$$



Note:

- Fourier Transform of impulse train is another impulse train
- Convolution with an impulse train is a shifting operation

Natural Sampling



If we multiply $x(t)$ by a train of rectangular pulses $x_p(t)$, we obtain a gated waveform that approximates the ideal sampled waveform, known as **natural sampling or gating**

$$x_s(t) = x(t)x_p(t)$$

$$= x(t) \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_s t}$$

$$X_s(f) = \mathfrak{F}[x(t)x_p(t)]$$

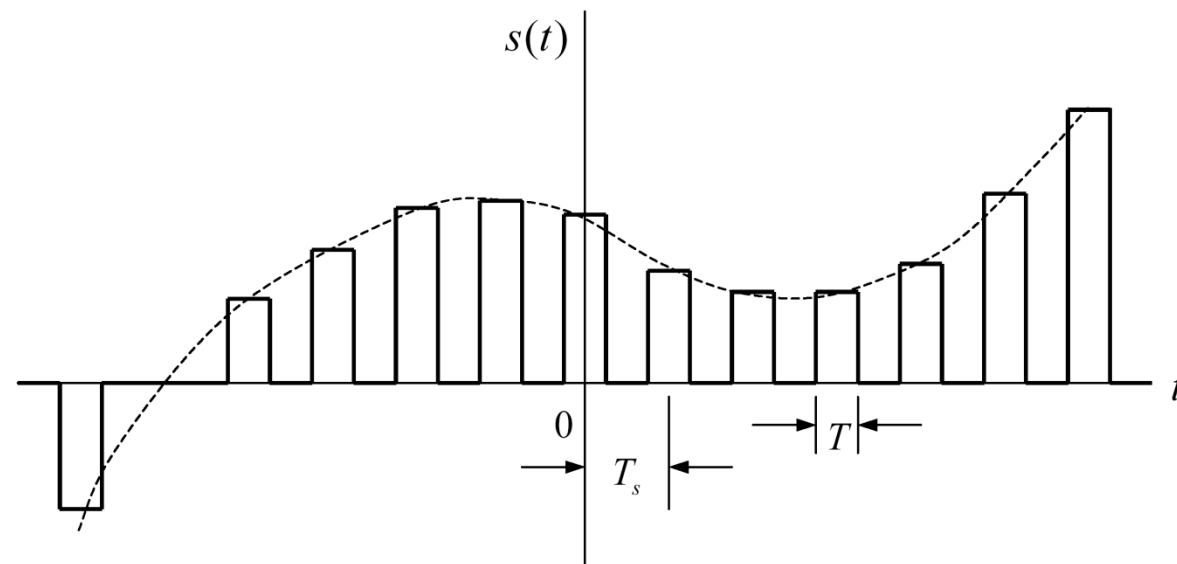
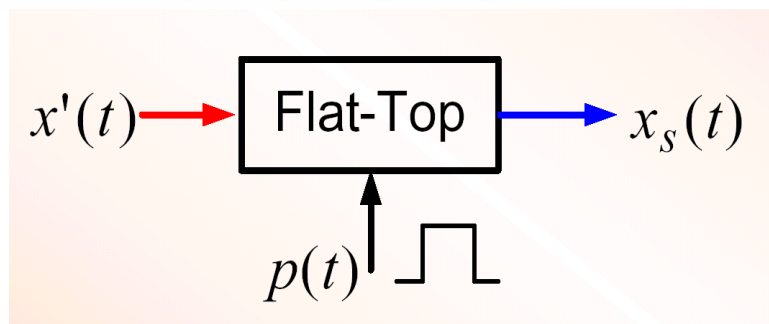
$$= \sum_{n=-\infty}^{\infty} c_n \mathfrak{F}[x(t)e^{j2\pi n f_s t}]$$

$$= \sum_{n=-\infty}^{\infty} c_n X[f - n f_s]$$

- Each pulse in $x_p(t)$ has width T_s and amplitude $1/T_s$
- The top of each pulse follows the variation of the signal being sampled
- $X_s(f)$ is the replication of $X(f)$ periodically every f_s Hz
- $X_s(f)$ is weighted by $C_n \leftarrow$ Fourier Series Coefficient
- The problem with a natural sampled waveform is that the tops of the sample pulses are not flat
- It is not compatible with a digital system since the amplitude of each sample has infinite number of possible values
- Another technique known as ***flat top sampling*** is used to alleviate this problem

Flat-Top Sampling

- Here, the pulse is held to a constant height for the whole sample period
- Flat top sampling is obtained by the convolution of the signal obtained after ideal sampling with a unity amplitude rectangular pulse, $p(t)$
- This technique is used to realize ***Sample-and-Hold*** (S/H) operation
- In S/H, input signal is continuously sampled and then the value is held for as long as it takes for the A/D to acquire its value
- Effect of the hold operation is the significant attenuation of the higher frequency spectral replicates.



$$x'(t) = x(t)\delta(t)$$

$$x_s(t) = x'(t) * p(t)$$

$$= p(t) * x(t)\delta(t) = p(t) * \left[x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right]$$

- Taking the Fourier Transform will result to

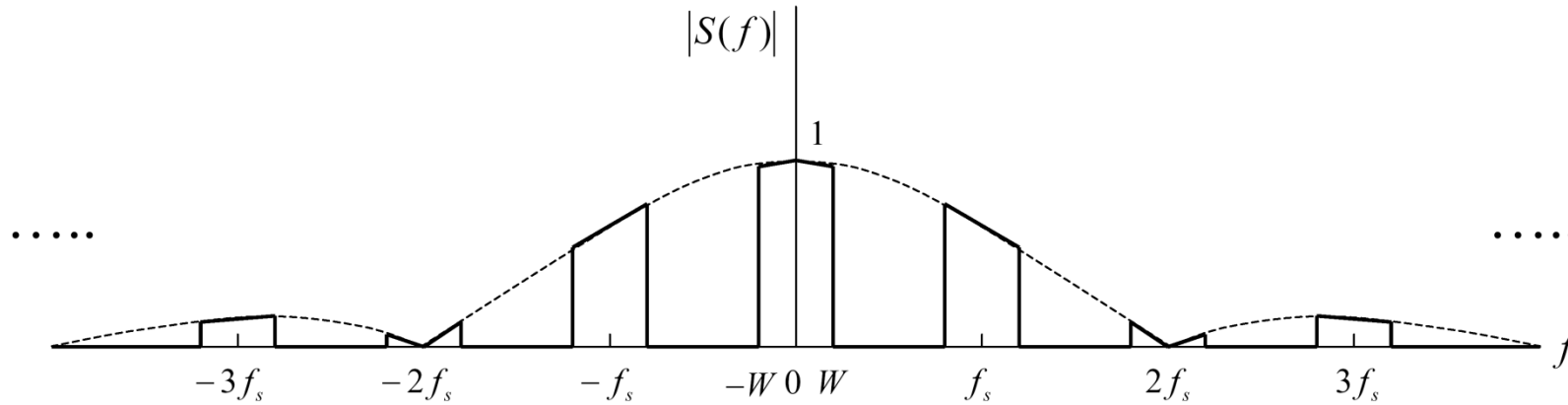
$$X_s(f) = \mathfrak{F}[x_s(t)]$$

$$= P(f) \mathfrak{F} \left[x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right]$$

$$= P(f) \mathfrak{F} \left[X(f) * \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \right]$$

$$= P(f) \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

where $P(f)$ is a *sinc* function

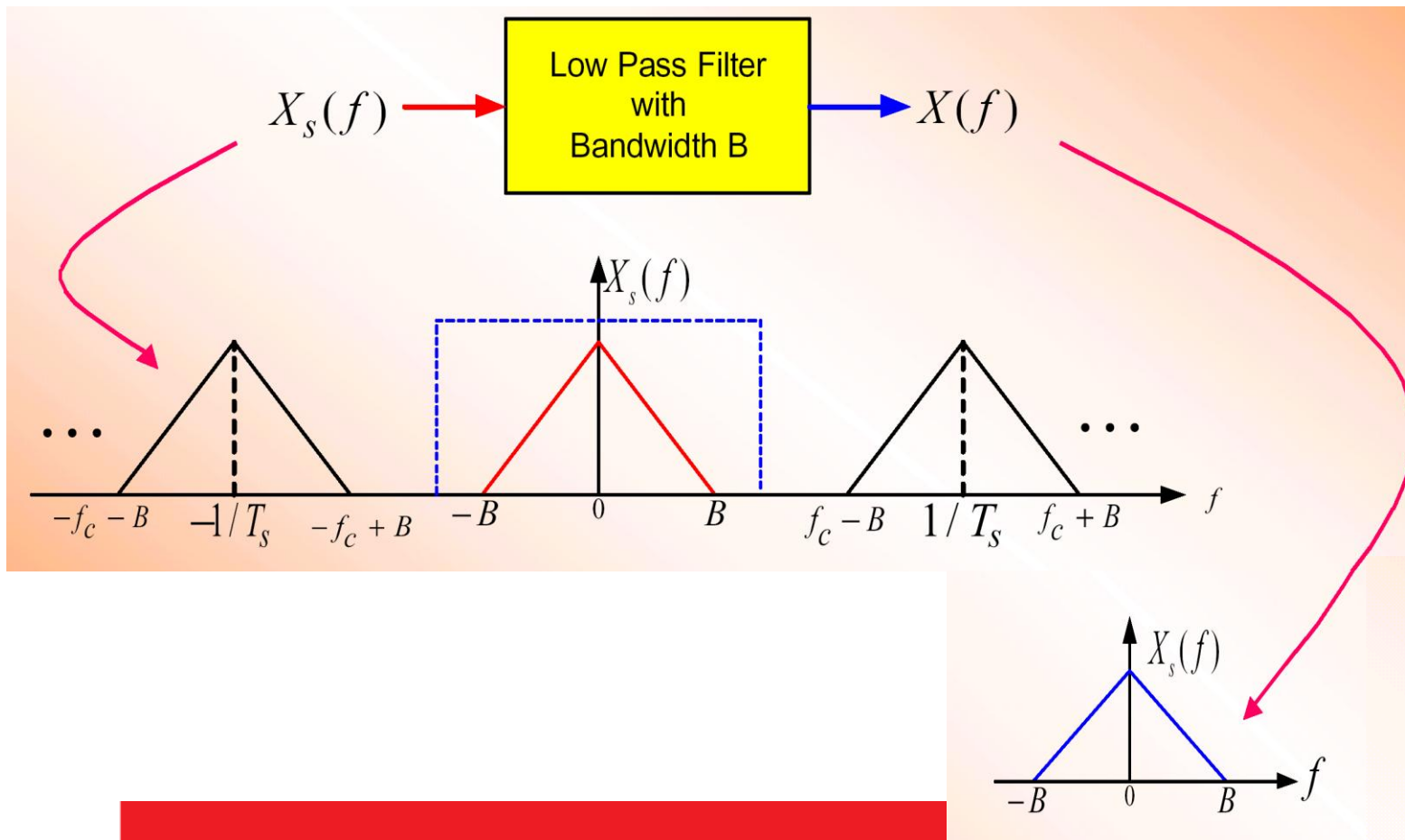


Flat top sampling (Frequency Domain)

- Flat top sampling becomes identical to ideal sampling as the width of the pulses become shorter

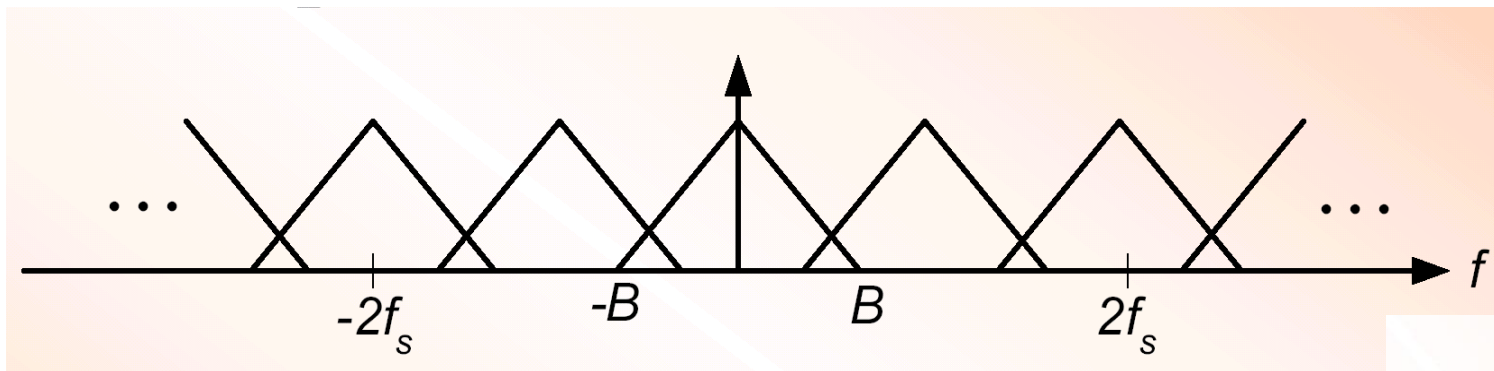
Recovering the Analog Signal

- One way of recovering the original signal from sampled signal $X_s(f)$ is to pass it through a Low Pass Filter (LPF) as shown below

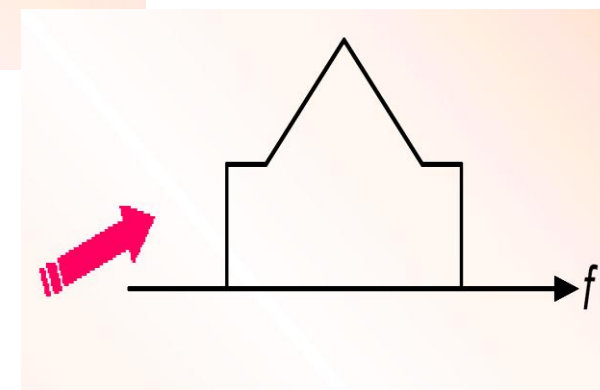


■ Undersampling and Aliasing

- If the waveform is **undersampled** (i.e. $f_s < 2B$) then there will be **spectral overlap** in the sampled signal

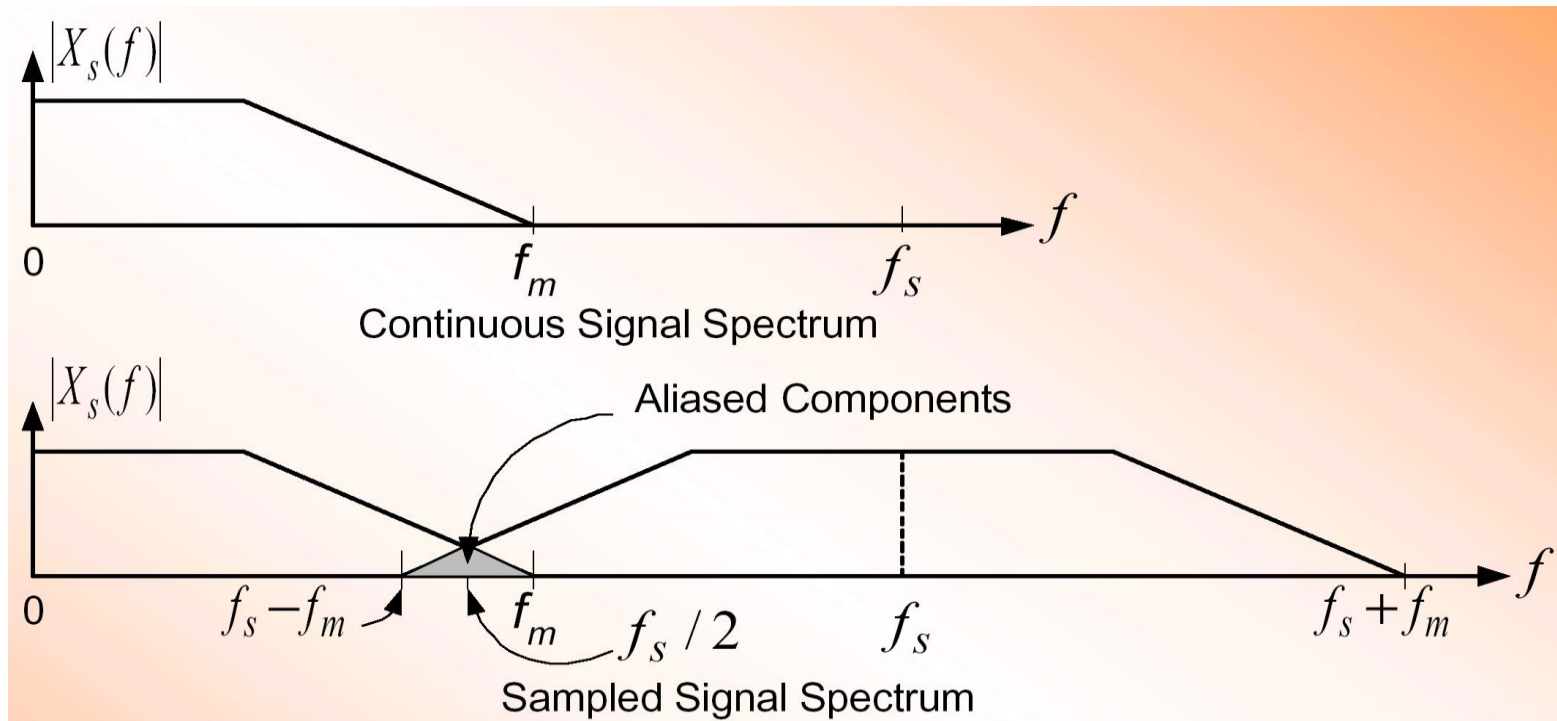


- The signal at the output of the filter will be different from the original signal spectrum



This is the outcome of **aliasing**!

- This implies that whenever the sampling condition is not met, an irreversible overlap of the spectral replicas is produced

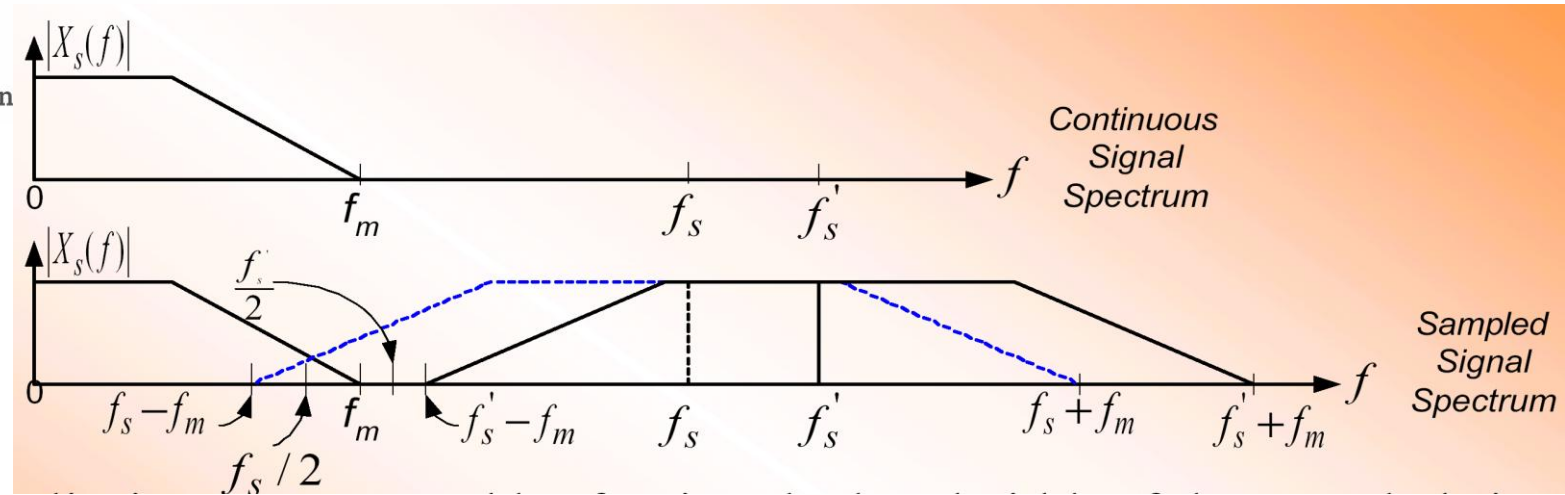


- This could be due to:
 1. $x(t)$ containing higher frequency than were expected
 2. An error in calculating the sampling rate
- Under normal conditions, under sampling of signals causing aliasing is not recommended



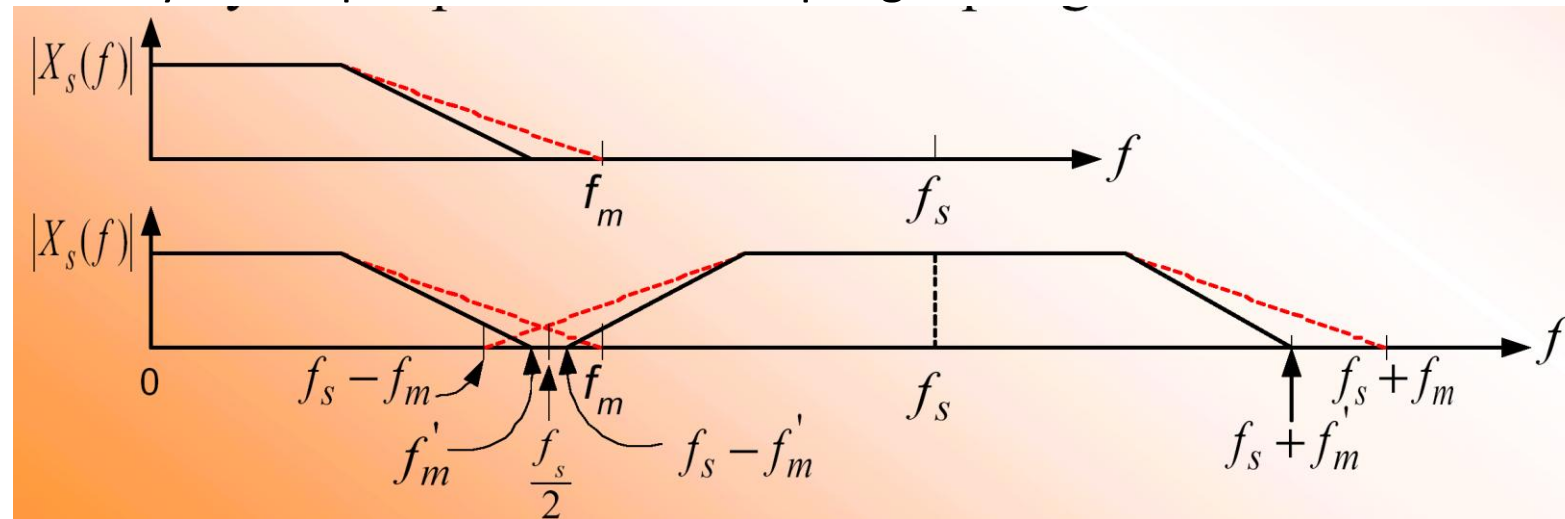
Solution 1: Anti-Aliasing Analog Filter

- ❑ All physically realizable signals are not completely bandlimited
- ❑ If there is a significant amount of energy in frequencies above half the sampling frequency ($f_s/2$), aliasing will occur
- ❑ Aliasing can be prevented by first passing the analog signal through an **anti-aliasing** filter (also called a **prefilter**) before sampling is performed
- ❑ The anti-aliasing filter is simply a LPF with cutoff frequency equal to half the sample rate



Case 2

- Aliasing is prevented by forcing the bandwidth of the sampled signal to satisfy the requirement of the Sampling Theorem



Case 1

■ **Solution 2: Over Sampling and Filtering in the Digital Domain**

- ❑ The signal is passed through a low performance (less costly) analog low-pass filter to limit the bandwidth.
- ❑ Sample the resulting signal at a high sampling frequency.
- ❑ The digital samples are then processed by a high performance digital filter and down sample the resulting signal.

- The alias frequency is given by

$$f_a = f_s / 2 - (f - f_s / 2) = f_s - f$$



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Example : A 5.5 kHz tone is sampled at 8 KHz . Find the alias frequency generated.



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Example : A 5.5 kHz tone is sampled at 8 KHz . Find the alias frequency generated.



Solution :

Here,

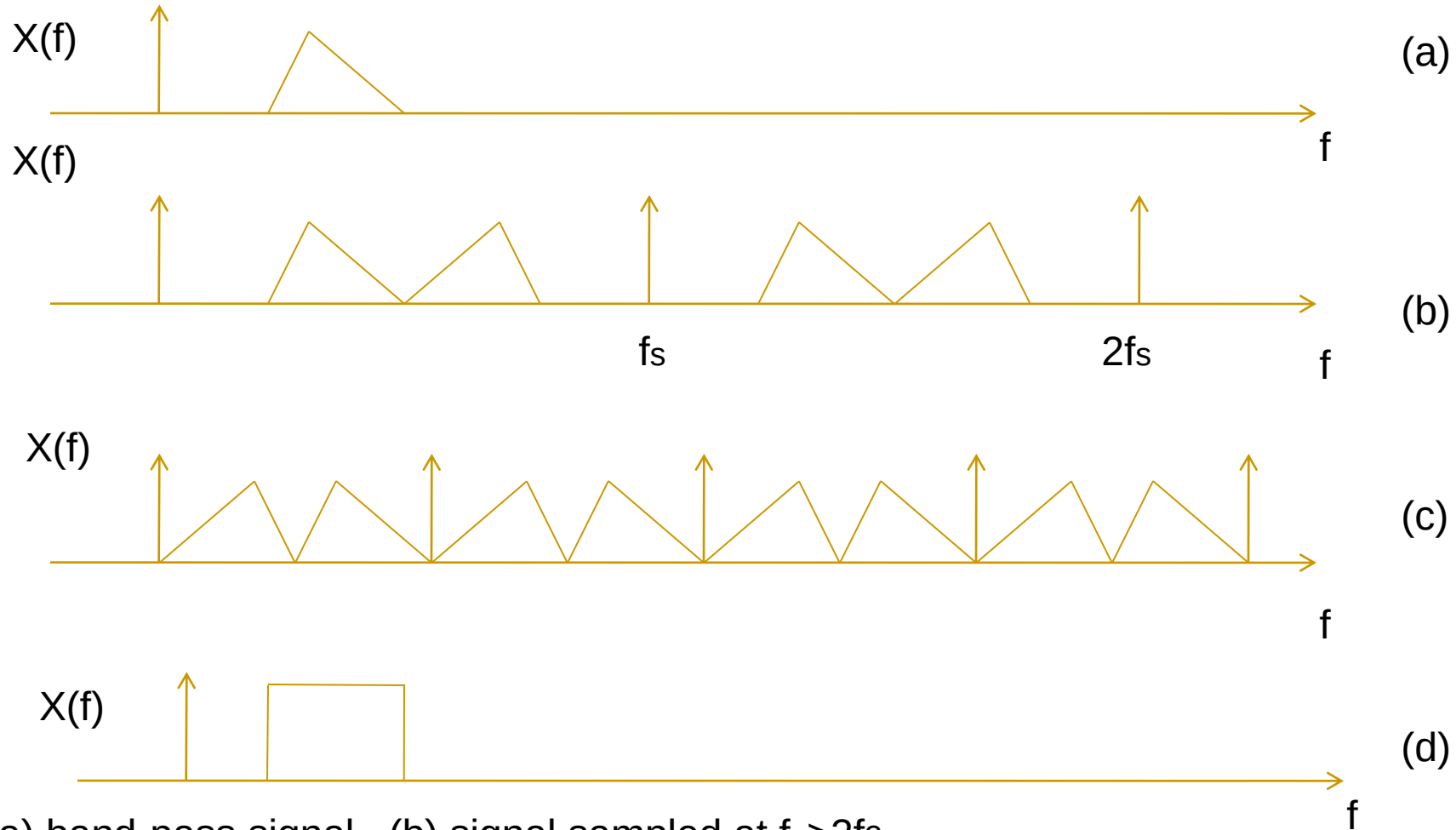
$$f = 5.5 \text{ kHz} \quad \text{and} \quad f_s = 8 \text{ kHz}$$

From

$$f_a = f_s / 2 - (f - f_s / 2) = f_s - f$$

$$f_a = 8000 / 2 - (5500 - 8000 / 2) = 8000 - 5500 = 2500 = 2.5 \text{ kHz}$$

Sampling of Bandpass Signals



- A band-pass signal occupies a frequency band from f_1 to f_2 .
- It has one sided spectrum as shown in figure (a).
- Sampling a band-pass signal at Nyquist rate spectrum is shown in fig (b).
- Clearly there are gaps in this spectrum.
- To avoid the spectrum overlap we can reduce the sampling frequency.
- Sampling frequency so arranged avoid overlapping of the spectrum fig(c).
- Minimum sampling frequency rate for a band-pass signal from the f_1 to f_2 , with bandwidth $B=f_2-f_1$ is given by

- $\text{Min } f_s = 2f_2 / N$ where N is integer part of the ratio f_2/B

- FDM signals and sub-band signals used in speech coding are examples of band-pass signals.



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Example : a FM signal at 10.7 MHz IF needs to be digitized for demodulation in a digital domain. If the bandwidth of the signal is 200 kHz, find the minimum usable sampling frequency

Example : a FM signal at 10.7 MHz IF needs to be digitized for demodulation in a digital domain. If the bandwidth of the signal is 200 kHz, find the minimum usable sampling frequency

Solution :

- The Bandwidth $B = f_2 - f_1 = 200$ kHz.
- The center frequency , $f_{IF} = (f_1 + f_2)/2 = 10,700$ kHz.
- Therefore the highest frequency $f_2 = 10,700 + 100 = 10,800$ kHz
- $N = f_2/B = 10,800/200 = 54$
- Minimum sampling frequency $f_s = 2 \cdot f_2/N = 2 \cdot (10,800 / 54) = 400$ kHz



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Example : A triangular waveform with 10 ms period is to be digitized. If the waveform fidelity is to be maintained up to its 10th harmonic, what should be the sampling frequency ?

Example : A triangular waveform with 10 ms period is to be digitized. If the waveform fidelity is to be maintained up to its 10th harmonic, what should be the sampling frequency ?

Solution :

The fundamental frequency of the triangular waveform is

$$f_0 = 1/T = 1 / (10 \times 10^{-3}) = 100 \text{ Hz}$$

The 10th harmonic is , therefore

$$f_m = 10 * f_0 = 10 \times 100 = 1000 \text{ Hz}$$

Hence the sampling frequency is

$$f_s = 2 \times 1000 = 2000 \text{ Hz}$$

Summary Of Sampling

- ❑ **Ideal Sampling**
(or Impulse Sampling)

$$\begin{aligned}x_s(t) &= x(t)x_\delta(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \\&= \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)\end{aligned}$$

- ❑ **Natural Sampling**
(or Gating)

$$x_s(t) = x(t)x_p(t) = x(t) \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_s t}$$

- ❑ **Flat-Top Sampling**

$$x_s(t) = x'(t) * p(t) = \left[x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right] * p(t)$$

- ❑ **For all sampling techniques**

- ❑ If $f_s > 2B$ then we can recover $x(t)$ **exactly**
- ❑ If $f_s < 2B$) **spectral overlapping** known as **aliasing** will occur

Practical Sampling Rates

■ Speech

- Telephone quality speech has a bandwidth of 4 kHz (actually 300 to 3300Hz)
- Most digital telephone systems are sampled at 8000 samples/sec

■ Audio:

- The highest frequency the human ear can hear is approximately 15kHz
- CD quality audio are sampled at rate of 44,000 samples/sec

■ Video

- The human eye requires samples at a rate of at least 20 frames/sec to achieve smooth motion

The signals $x_1(t) = 10 \cos(100\pi t)$ and $x_2(t) = 10 \cos(50\pi t)$ are both sampled at $t = nT_s$ where $n = 0, \pm 1, \pm 2, \dots$ and $f_s = 25$ samples/second. Show that the two sequences of the sampler thus obtained are identical. What is this phenomenon called ?



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The signal $x(t) = 20 + 20 \sin(500\pi t + 30^\circ)$ is to be sampled periodically and reproduced from the samples

- a) Find the maximum allowable time interval between the samples
- b) How many sample values need to be stored in order to reproduce 1 sec of this waveform, if sampled according to the result in (a)
- c) Determine and sketch spectrum of sampled signal when sampling frequency $f_s = 750$ Hz



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ASSIGNMENT

Example 1:

- Consider the analog signal $x(t)$ given by

$$x(t) = 3\cos(50\pi t) + 100\sin(300\pi t) - \cos(100\pi t)$$

- What is the Nyquist rate for this signal?

Example 2:

- Consider the analog signal $x_a(t)$ given by

$$x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + \cos 12000\pi t$$

- What is the Nyquist rate for this signal?
- What is the discrete time signal obtained after sampling, if $f_s = 5000$ samples/s.
- What is the analog signal $x(t)$ that can be reconstructed from the sampled values?

Coding

Coding

Assigning binary code to each step

Binary code is represented as Pulses

No pulse 0

Pulse 1

- There are many types of waveforms. Eg:PCM, PAM, PWM tc..Why? → performance criteria!
- Each line code type have merits and demerits
- The choice of waveform depends on operating characteristics of a system such as:
 - ❑ Modulation-demodulation requirements
 - ❑ Bandwidth requirement
 - ❑ Synchronization requirement
 - ❑ Receiver complexity, etc.,

Source vs. Channel vs. Line Coding

- **Source coding**: eliminating redundancy in order to make efficient use of storage space and/or transmission channels
 - Huffman coding/ Morse code
- **Channel coding**: a pre-transmission mapping applied to a digital signal or file, usually designed to make error-correction possible
 - Parity check / Hamming code / Reed-Soloman code
- **Line coding**: performed to adapt the transmitted signal to the (electrical) characteristics of a transmission channel
- Order: source coding -> channel coding -> line coding

Line Coding

- It is the process of converting digital data into digital signal
- It converts sequence of bits into digital signal.

Goals of Line Coding (qualities to look for)

- A line code is designed to meet one or more of the following goals:

- **Self-synchronization**

- The ability to recover timing from the signal itself
 - That is, self-clocking (self-synchronization) - ease of clock lock or signal recovery for symbol synchronization
- Long series of ones and zeros could cause a problem

- **Low probability of bit error**

- Receiver needs to be able to distinguish the waveform associated with a *mark* from the waveform associated with a *space*
- BER performance
 - relative immunity to noise
- Error detection capability
 - enhances low probability of error

- ❑ **Spectrum Suitable for the channel**
 - Spectrum matching of the channel
 - ❑ e.g. presence or absence of DC level
 - In some cases DC components should be avoided
 - The transmission bandwidth should be minimized
- ❑ **Power Spectral Density**
 - Particularly its value at zero
 - ❑ PSD of code should be negligible at the frequency near zero
- ❑ **Transmission Bandwidth**
 - Should be as small as possible
- ❑ **Transparency**
 - The property that any arbitrary symbol or bit pattern can be transmitted and received, i.e., all possible data sequence should be faithfully reproducible

What is Line Coding?

- Mapping of binary information sequence into the digital signal that enters the channel
 - Ex. “1” maps to +A square pulse; “0” to –A pulse
- Line code selected to meet system requirements:
 - *Transmitted power*: Power consumption = \$
 - *Bit timing*: Transitions in signal help timing recovery
 - *Bandwidth efficiency*: Excessive transitions wastes bw
 - *Low frequency content*: Some channels block low frequencies
 - long periods of +A or of –A causes signal to “droop”
 - Waveform should not have low-frequency content
 - *Error detection*: Ability to detect errors helps
 - *Complexity/cost*: Is code implementable in chip at high speed?

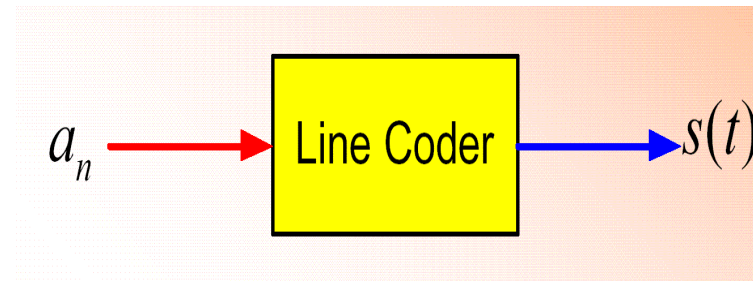
Line Coder

- The input to the line encoder is the output of the A/D converter or a sequence of values a_n that is a function of the data bit
- The output of the line encoder is a waveform:

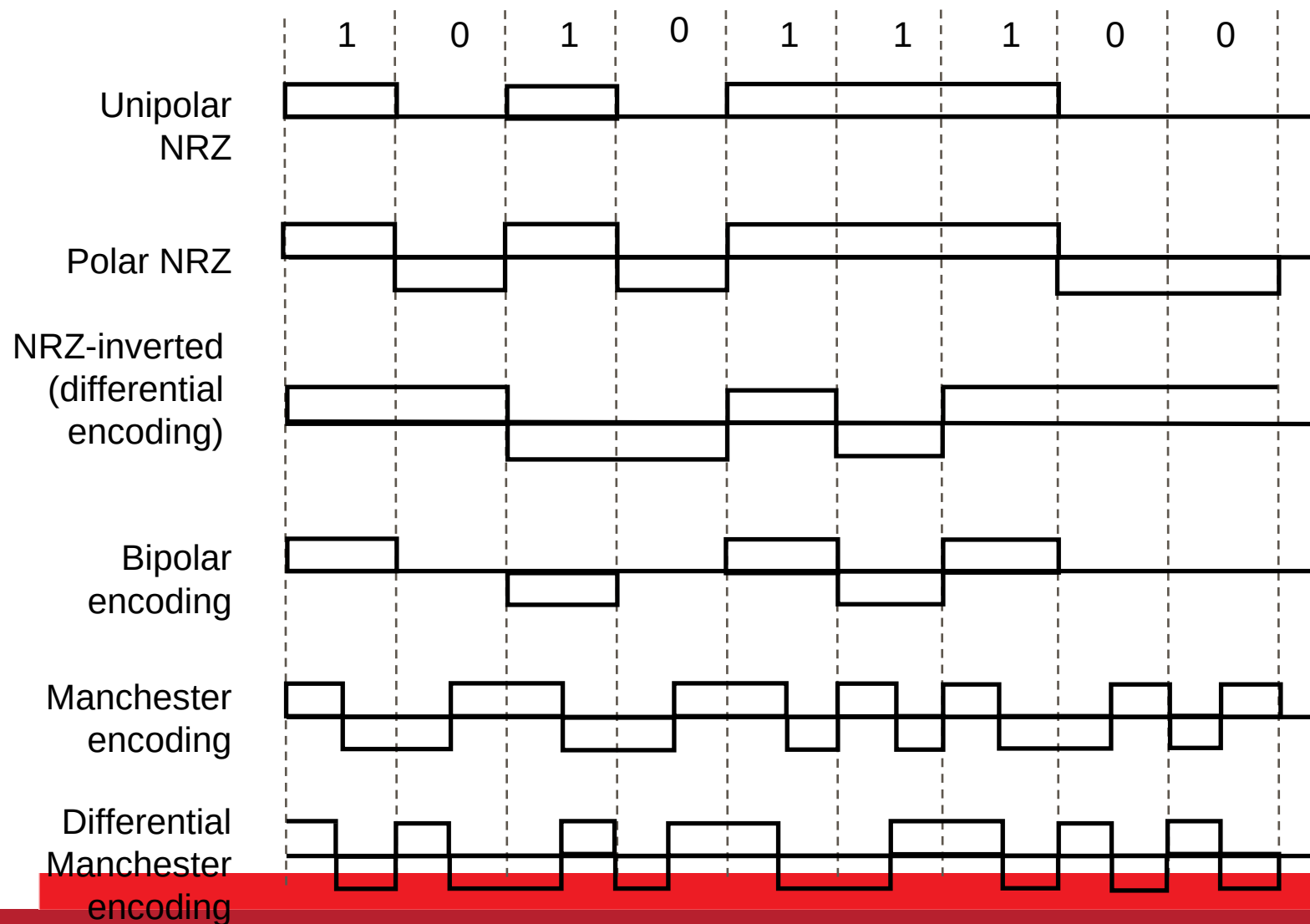
$$s(t) = \sum_{n=-\infty}^{\infty} a_n f(t - nT_b)$$

where $f(t)$ is the pulse shape and T_b is the bit period ($T_b = T_s/n$ for n bit quantizer)

- This means that each line code is described by a symbol mapping function a_n and pulse shape $f(t)$
- Details of this operation are set by the type of line code that is being used

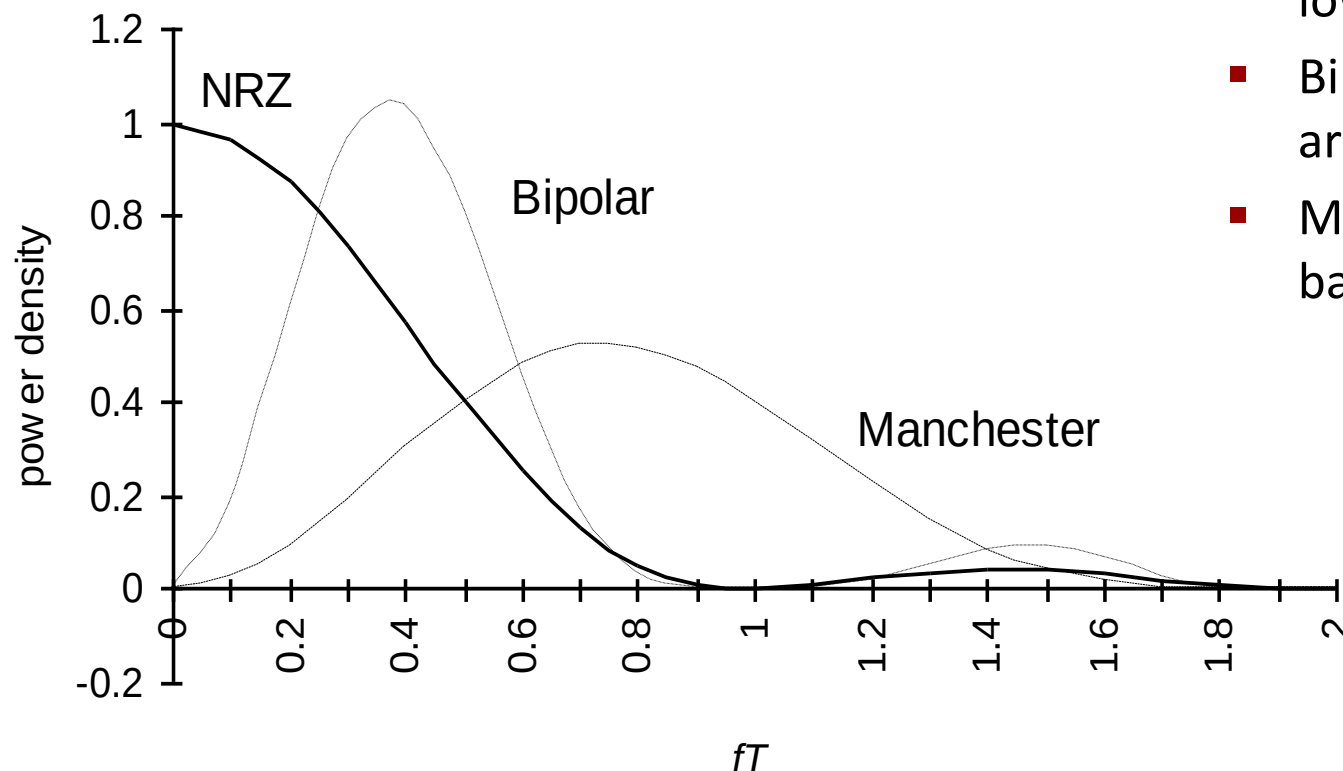


Line coding examples



Spectrum of Line codes

- Assume 1s & 0s independent & equiprobable

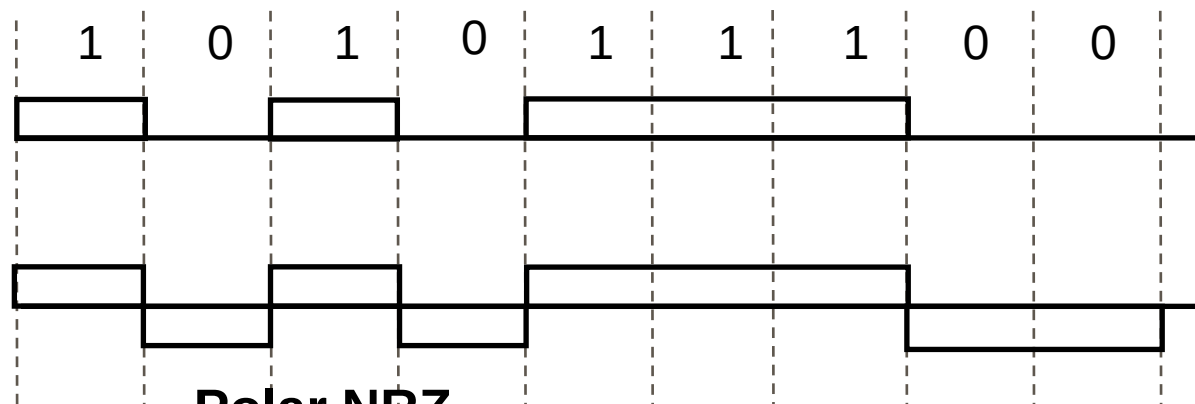


- NRZ has high content at low frequencies
- Bipolar tightly packed around $f=1/(2T)$ or $0.5/T$
- Manchester wasteful of bandwidth

Unipolar & Polar Non-Return-to-Zero (NRZ)

Unipolar NRZ

Polar NRZ



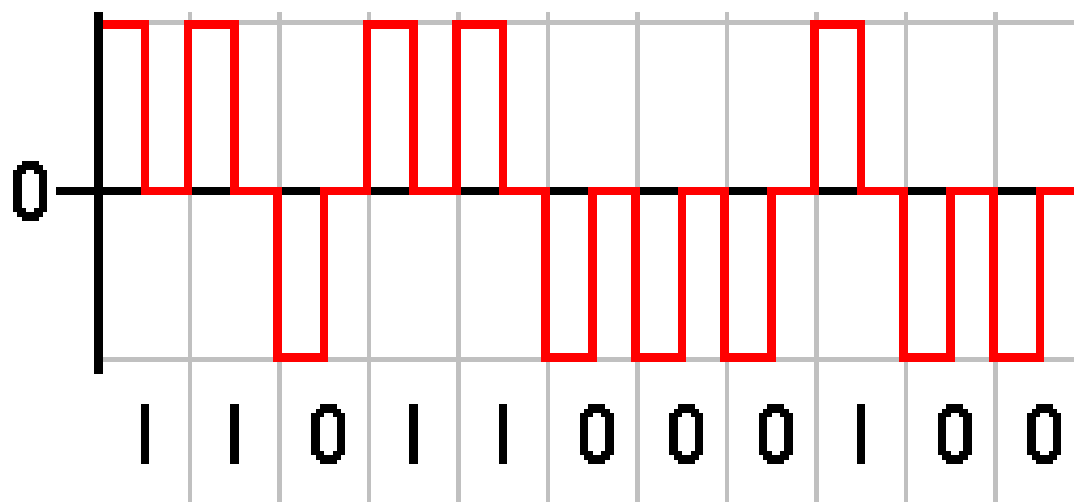
Unipolar NRZ

- “1” maps to +A pulse
- “0” maps to no pulse
- High Average Power
 $0.5 \cdot A^2 + 0.5 \cdot 0^2 = A^2/2$
- Long strings of A or 0
 - Poor timing
 - Low-frequency content
- Simple

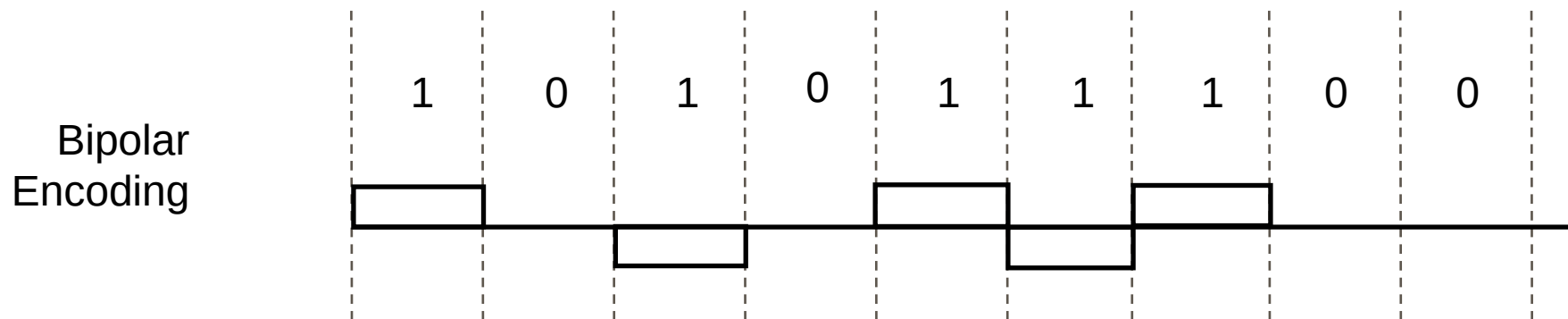
Polar NRZ

- “1” maps to +A/2 pulse
- “0” maps to -A/2 pulse
- Better Average Power
 $0.5 \cdot (A/2)^2 + 0.5 \cdot (-A/2)^2 = A^2/4$
- Long strings of +A/2 or -A/2
 - Poor timing
 - Low-frequency content
- Simple

Return-to-Zero (RZ)



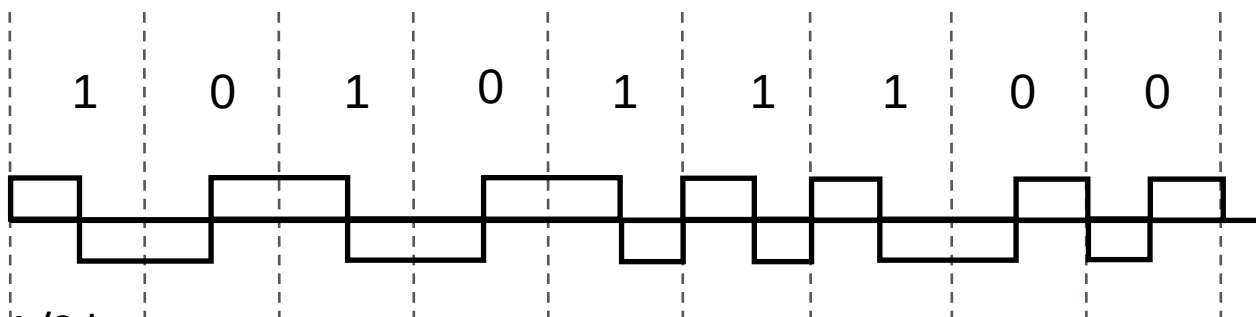
Bipolar Code



- Three signal levels: $\{-A, 0, +A\}$
- “1” maps to +A or -A in alternation
- “0” maps to no pulse
 - Every +pulse matched by -pulse so little content at low frequencies
- String of 1s produces a square wave
 - Spectrum centered at around $f=1/(2T)$ or $0.5/T$
- Long string of 0s causes receiver to lose synchronization
- Zero-substitution codes are needed

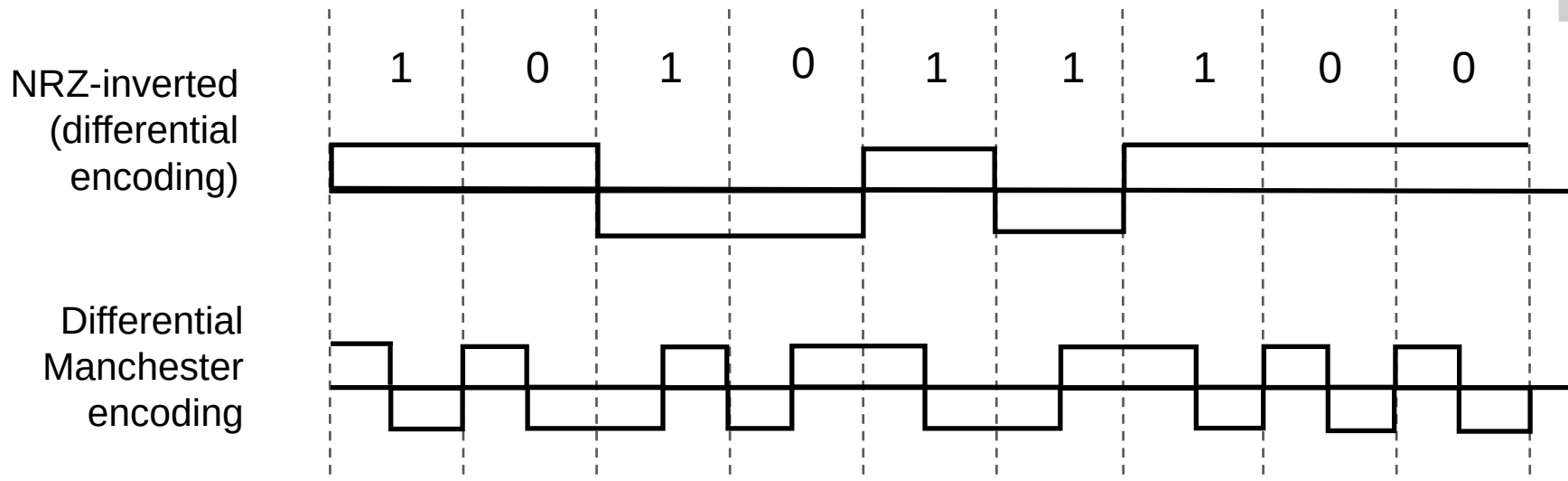
Manchester code & $mBnB$ codes

Manchester
Encoding



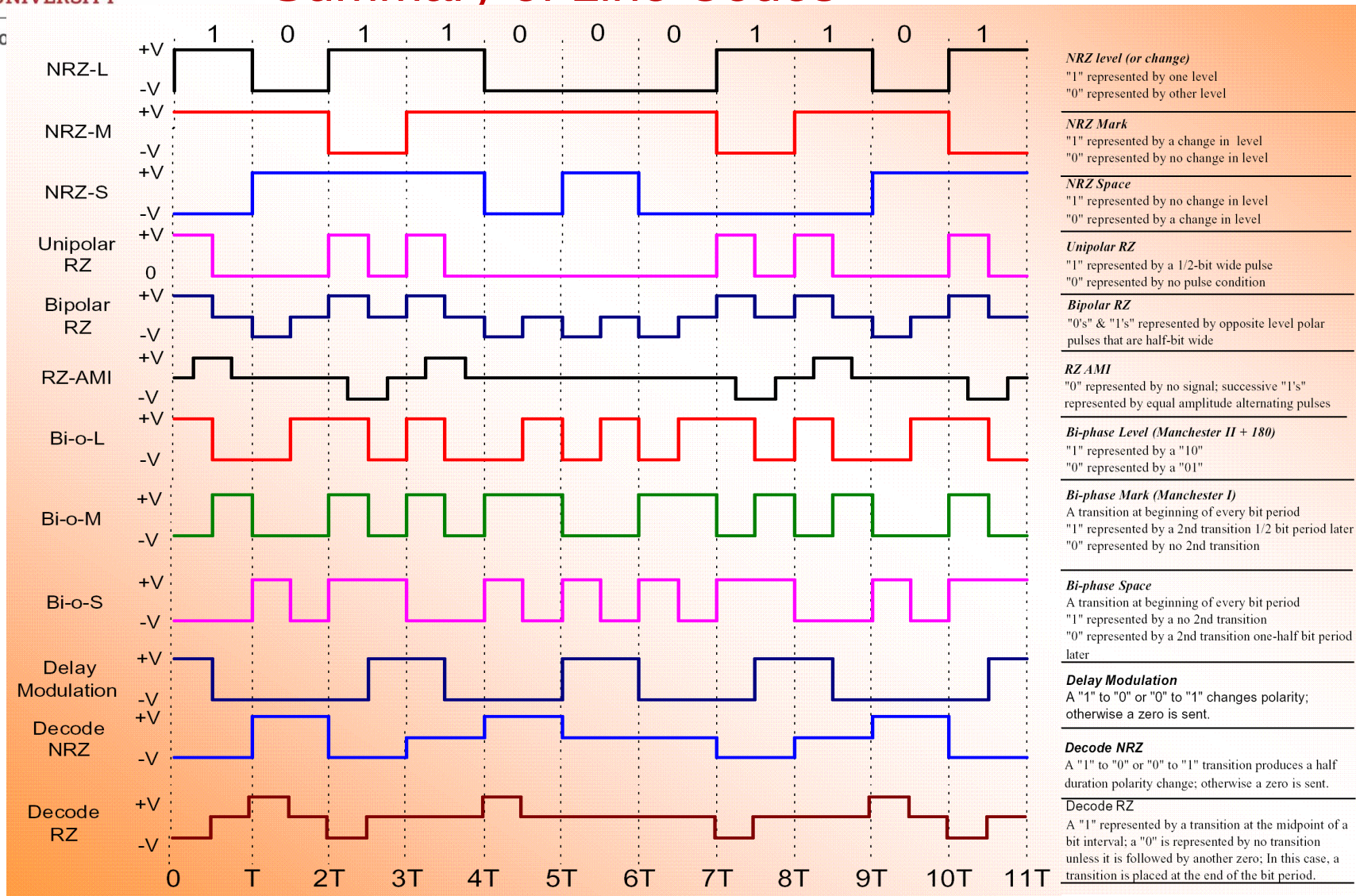
- “1” maps into $A/2$ first $T/2$, $-A/2$ last $T/2$
- “0” maps into $-A/2$ first $T/2$, $A/2$ last $T/2$
- Every interval has transition in middle
 - Timing recovery easy
 - Uses double the minimum bandwidth
- Simple to implement
- Used in 10-Mbps Ethernet & other LAN standards
- $mBnB$ line code
- Maps block of m bits into n bits
- Manchester code is 1B2B code
- 4B5B code used in FDDI LAN
- 8B10b code used in Gigabit Ethernet
- 64B66B code used in 10G Ethernet

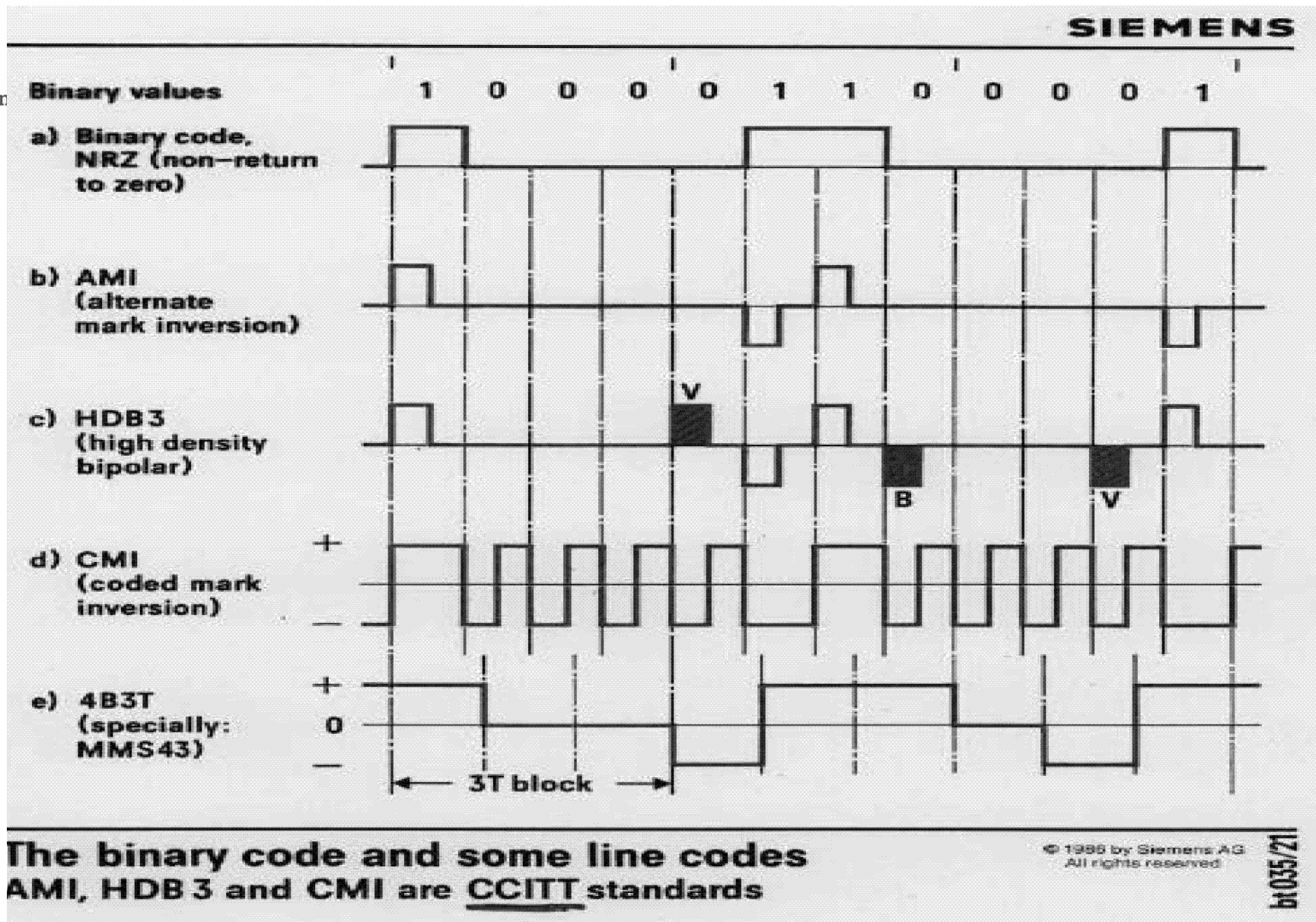
Differential Coding



- Errors in some systems cause transposition in polarity, +A become -A and vice versa
- All subsequent bits in Polar NRZ coding would be in error
- Differential line coding provides robustness to this type of error
- “1” mapped into transition in signal level
- “0” mapped into no transition in signal level
- Also used along with Manchester coding

Summary of Line Codes





Comparison of Line Codes

- Self-synchronization
 - ❑ Manchester codes have built in timing information because they always have a zero crossing in the center of the pulse
 - ❑ Polar RZ codes tend to be good because the signal level always goes to zero for the second half of the pulse
 - ❑ NRZ signals are not good for self-synchronization
- Error probability
 - ❑ Polar codes perform better (are more energy efficient) than Unipolar or Bipolar codes
- Channel characteristics
 - ❑ We need to find the power spectral density (PSD) of the line codes to compare the line codes in terms of the channel characteristics

Comparison of Line Codes

- Different pulse shapes are used
 - ❑ to control the spectrum of the transmitted signal (no DC value, bandwidth, etc.)
 - ❑ guarantee transitions every symbol interval to assist in symbol timing recovery

1. Power Spectral Density of Line Codes

- After line coding, the pulses may be filtered or shaped to further improve their properties such as
 - ❑ Spectral efficiency
 - ❑ Immunity to Intersymbol Interference
- Distinction between Line Coding and Pulse Shaping is not easy

2. DC Component and Bandwidth

- DC Components
 - ❑ Unipolar NRZ, polar NRZ, and unipolar RZ all have DC components
 - ❑ Bipolar RZ and Manchester NRZ do not have DC components



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Information Theory

Source coding

What is information?

Information Theory

Information is linked with the element of surprise or uncertainty.

In terms of probability.

The more probable some event to occur the less information related to its occurrence

The less probable some event to occur the more information

Source coding

How we can measure the information?

Measure of Information

Given a digital source with N possible outcomes

“messages”, the information sent from the digital source when the jth message is transmitted is given by the following equation

$$I = \log[(1/P_j)] / \log 2 \quad [\text{bits}]$$

Source coding

Find the information content of a message that takes one hour of 4 possible outcomes equally likely.

Probability of each outcome $P = 1/0.25$

$$I = \log 1/p = [\log 1/(1/0.25)]/[\log 2] =$$

Source coding

Entropy : H

The entropy of a discrete source S is the average amount of information (or uncertainty) associated with that source.



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NEXT PROBLEM



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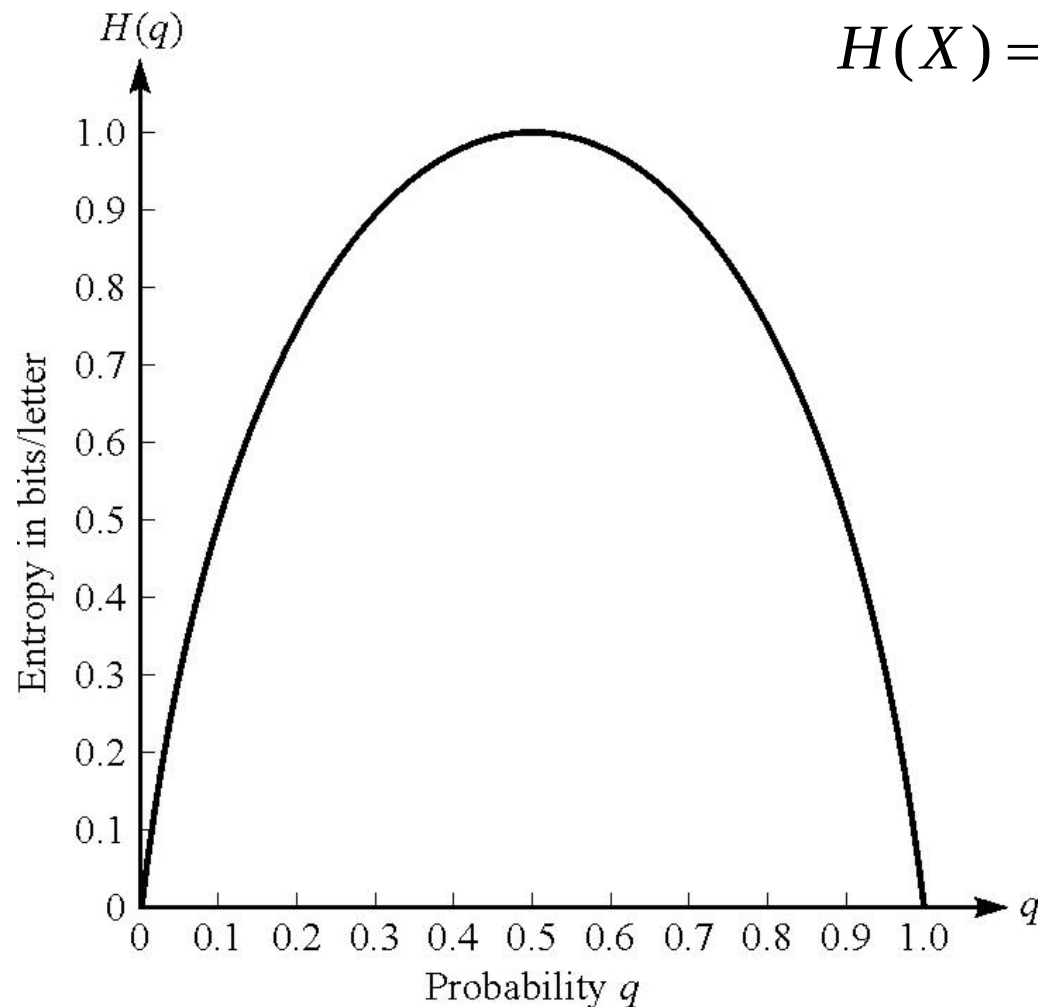


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Binary entropy function



$$H(X) = -p \log p - (1-p) \log(1-p)$$

$H=0$: no uncertainty

$H=1$: most uncertainty

1 bit for binary information

Channel Capacity Theorem

- $X(t)$ is bandlimited to B Hz
- If random process is limited to interval, $-B \leq f \leq B$, is sampled at a rate $\geq 2B$ samples /second then number of samples $n = 2BT$
- Channel used n times for transmission of n samples of process $X(t)$ in T seconds, channel capacity per unit time is (n/T) times.
- $C = B \log_2 \left(1 + \frac{P}{N_0 B} \right) \text{ bits/sec Hz}$
where P is average transmitted power, awgn of power spectral density $N_0/2$

$$C = B \log_2 \left(1 + \frac{P}{N_0 B} \right) \text{ bits/sec Hz}$$

N_0 is power spectral density = Noise Power/ Bandwidth

- $= B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/sec}$
- **Ex1:** Find capacity of Gaussian channel of bandwidth 4KHz and noise PSD 10^{-9} W/Hz when signal energy is a) 0.1J b) 0.001J c) how does channel capacity changes in b if bandwidth is increased to 10KHz?
- $C = B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/sec}$

$$\begin{aligned} \text{a) } C &= B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/sec} \\ &= 4000 \log_2 \left(1 + \frac{0.1}{1 * 4000 * 10^{-9}} \right) \text{ bits/sec} \\ &= *** \times 10^4 \text{ bit/sec} \\ &= 58438 \text{ bit/sec} \end{aligned}$$

$$\text{b) } C = **** \times 10^4 \text{ bit/sec}$$

c) If $B = 10000$,

$$\begin{aligned} C &= ***** \times 10^4 \text{ bit/sec} \\ &= 132878 \text{ bit/sec} \end{aligned}$$

Capacity is doubled by increasing bandwidth by 2.5 times.

EX2: For TV transmission, the required number of brightness level =16, pixels /picture frame = 10^6 , frames transmitted per second = 30 and SNR=30dB. Find minimum bandwidth required.

- 30dB = 1000
- Information /pixel = $\log_2 16 = 4$ bits/pixel
- Information /picture frame = 4×10^6 bits/frame
- Information rate = $30 \times 4 \times 10^6$ bits/frame

Channel capacity $C > 12 \times 10^7$ bits/frame

- $C = B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/sec}$

$$B \log_2 \left(1 + \frac{S}{N} \right) \geq 12 \times 10^7$$

$$B \log_2 (1 + 1000) \geq 12 \times 10^7$$

$$B \geq 1.212 \times 10^6$$

Minimum bandwidth = 12.12 MHz



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Huffman coding Optimum as it provides smallest avg codeword

- ❑ Huffman coding is a popular lossless Variable Length Coding (VLC) scheme, based on the following principles:
- ❑ Shorter code words are assigned to more probable symbols and longer code words are assigned to less probable symbols.
- ❑ No code word of a symbol is a prefix of another code word. This makes Huffman coding uniquely decodable.
- ❑ Every source symbol must have a unique code word assigned to it.

Huffman coding Optimum as it provides smallest avg codeword

- ❑ Huffman Coding is a famous Greedy Algorithm.
- ❑ It is used for the lossless compression of data.
- ❑ It uses variable length encoding.
- ❑ The code length of a character depends on how frequently it occurs in the given text.
- ❑ The character which occurs most frequently gets the smallest code.
- ❑ The character which occurs least frequently gets the largest code.

Huffman coding Optimum as it provides smallest avg codeword

There are two major steps in Huffman Coding-

- ❑ Building a Huffman Tree from the input characters.
- ❑ Assigning code to the characters by traversing the Huffman Tree.



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Huffman coding Optimum as it provides smallest avg codeword

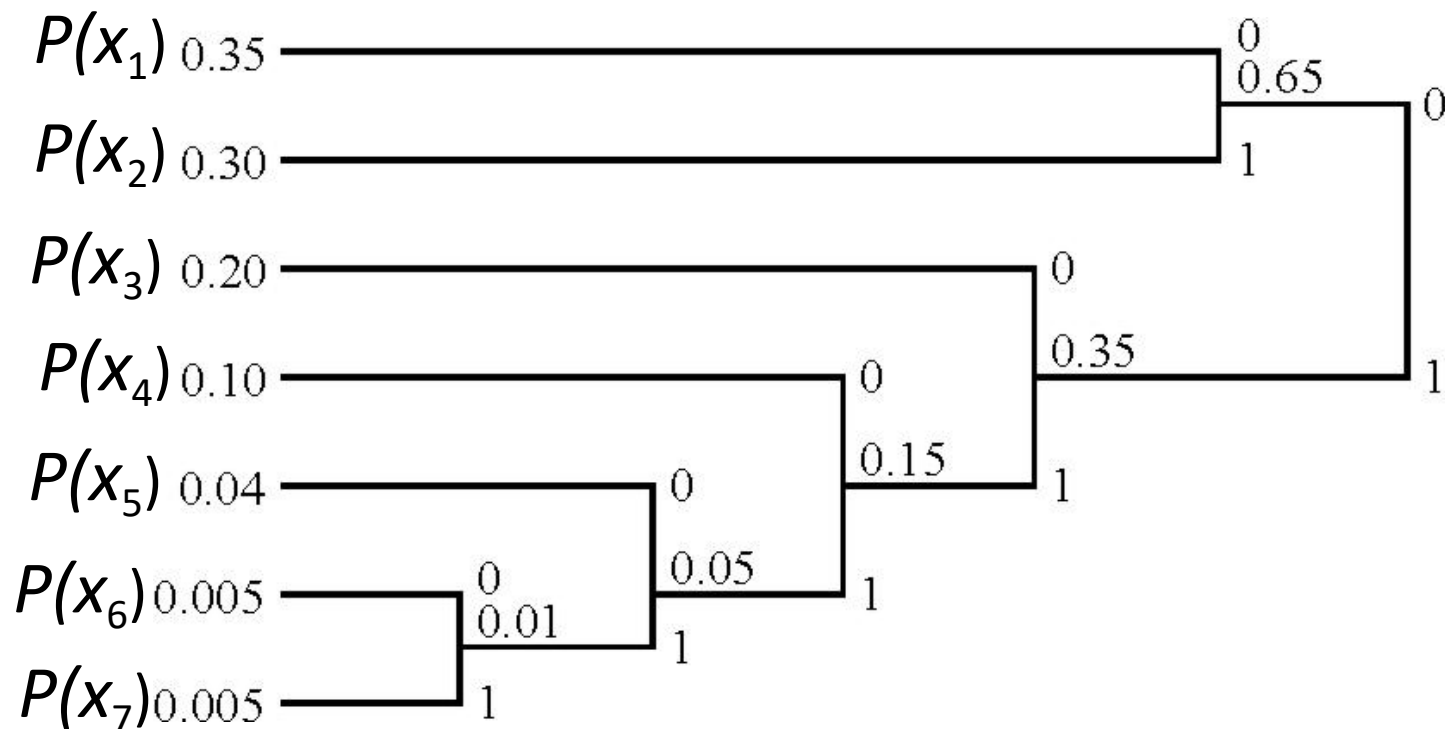
List the source symbols (msgs) in the order of decreasing probability.
Two sources with lowest probability are assigned numbers 0 and 1
These sources symbols are Combined to a new message as the sum of two original probabilities.
The probability of this new msg is placed in the list according to its value.
Repeat this procedure until we are left with only two source symbols for which a 0 and 1 are assigned .
The code of each original source symbol is found by working backward and tracing sequence of 0s and 1s assigned to that symbol as well as its successors.

Huffman coding example 1

- Consider the seven source symbols (msgs) of a discrete memoryless source and their probabilities. Follow the Huffman coding to find the codewords of each msg.

- | message | m1 | m2 | m3 | m4 | m5 | m6 | m7 |
|-------------|-----|-----|-----|------|-------|-------|------|
| probability | 0.1 | 0.2 | 0.3 | 0.04 | 0.005 | 0.005 | 0.35 |

Huffman coding algorithm



$$H(X) = 2.11$$

$R = 2.21$ bits per symbol

x_1	00
x_2	01
x_3	10
x_4	110
x_5	1110
x_6	11110
x_7	11111

Huffman coding example 1

- Consider the five source symbols (msgs) of a discrete memoryless source and their probabilities. Follow the Huffman coding to find the codewords of each msg.

- | message | m1 | m2 | m3 | m4 | m5 |
|-------------|-----|-----|-----|-----|-----|
| probability | 0.4 | 0.2 | 0.2 | 0.1 | 0.1 |

Huffman coding example 2

- Consider the five source symbols (msgs) of a discrete memoryless source and their probabilities. Follow the Huffman coding to find the code words of each msg.

- | message | m1 | m2 | m3 | m4 | m5 | m6 | m7 |
|-------------|------|------|-------|-------|-------|--------|--------|
| probability | 0.25 | 0.25 | 0.125 | 0.125 | 0.125 | 0.0625 | 0.0625 |



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Shannon-Fano Coding

Simplest variable length method. Less efficient than Huffman, but allows one to code symbol s_i with length l_i directly from probability p_i .

$$l_i = \lceil \log_r(1/p_i) \rceil$$

$$* \quad \left(\log_r \frac{1}{p_i} \right) \leq l_i < \left(\log_r \frac{1}{p_i} \right) + 1 \Rightarrow \frac{1}{p_i} \leq r^{l_i} < \frac{r}{p_i} \Rightarrow p_i \geq r^{-l_i} > \frac{p_i}{r}.$$

Summing this inequality over i :

$$\sum_{i=1}^q p_i = 1 \geq \sum_{i=1}^q \overbrace{r^{-l_i}}^K > \sum_{i=1}^q \frac{p_i}{r} = \frac{1}{r}$$

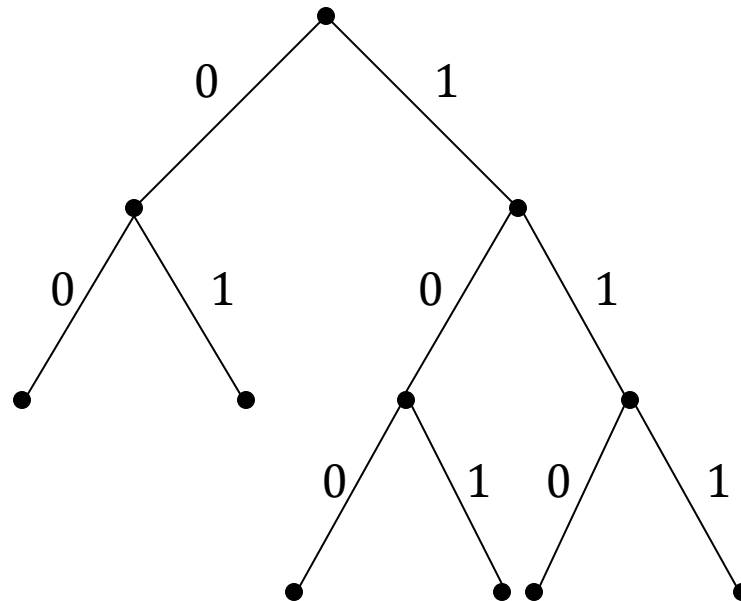
Kraft inequality is satisfied, therefore there is an instantaneous code with these lengths.

$$\text{Also, } H_r(S) = \sum_{i=1}^q p_i \log_r \frac{1}{p_i} \leq \underbrace{\sum_{i=1}^q p_i l_i}_L < H_r(S) + 1$$

by summing * multiplied by p_i

Example: p 's: $\frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}$ l 's: 2, 2, 3, 3, 3, 3 $K = 1$

$$H_2(S) = 2.5$$



$$L = 5/2$$



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1. Peter piper picked a peck of pickled peppers.
2. Better Botter bought some butter.
3. She sells seashells by the sea shore.
4. I scream , you scream, we all scream for ice cream.



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